Markovian Approximation of Long-Range Dependence in Implied Volatility Surfaces

Alfredo Diodati (VU Student Number: 2903270)

supervisor: prof. dr. A. Lucas
co-reader: prof. dr. F. Hamelink

June 29, 2026

Abstract

Fractional Integration and Random Multiplicative Cascades are the most widely used methods to model time series with Long Range Dependence, but are often challenging to use due to their computational requirements. Instead, this thesis presents a procedure that approximates Long Range Dependence parsimoniously. Unlike Fractional Integration, this procedure generates long-range State-Space models with a finite state dimension, and unlike multiplicative cascades it can be efficiently estimated by the Kalman Filter. The approximation is also adjusted for Score-Driven models, making it possible to use non-Gaussian features without simulation-based estimation of the parameters. The methodology is applied to Score-Driven and State-Space dynamic factor models for the implied volatility surface of the S&P 500. Monte Carlo studies demonstrate that the approximation procedure effectively replicates true Long Range Dependence characteristics, outperforming short-memory models in point forecasts at multiple horizons. When tested on real option data it is found that both the approximating and true models have similar forecasting performance to short-memory models at different horizons.

Key words: Long Range Dependence, Implied Volatility, Score-Driven model, State-Space model.

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1 Introduction

Long Range Dependence is a property of a stationary time series whereby its autocovariances decay slowly as the lag between observations grows. In a short-memory process the dependence between two observations vanishes geometrically with the lag separating them; in a Long Range Dependence process it vanishes far more slowly, often at a power-law rate. Writing $\gamma(\tau)$ for the autocovariance at lag τ, this slow decay takes the form $\gamma(\tau) = c\tau^{-\eta}$ for a scaling exponent $\eta \in (0, 2)$ and a constant $c > 0$.

The standard means of generating this power law decay is fractional integration (Granger and Joyeux, 1980; Hosking, 1981; Baillie et al., 1996), which represents the current value as a weighted average of *all* past values with slowly declining weights. While keeping track of all past values is feasible in such standard models, which are often univariate and observation driven, it makes computations intensive in multivariate and/or parameter driven models. For example, linear Gaussian fractionally integrated models can be cast into the State-Space framework and be estimated through the Kalman filter (see Durbin and Koopman (2012)), but their computation is extensive due to the increasing size of the state. This drastically increases computational and computer-memory requirements, which is why practitioners often use truncation procedures (see Chan and Palma (1998)). Further computational requirements arise when the State-Space is nonlinear or non-Gaussian, and sequential Monte Carlo methods such as the particle filter are required (see Chronopoulou and Spiliopoulos (2018)).

An alternative procedure that generates Long Range Dependence Processes is the use of Random Multiplicative Cascades as in Calvet and Fisher (2002) and Calvet and Fisher (2004). In particular, these processes generate Long Memory features by adding the product of a large number of independent Poisson Hidden Markov Models (HMM) to the state equation. Not only does this methodology generate the power law structure in the autocovariance but also in all moments of the process, a feature often referred to as 'multifractality.' However, this approach also has computational[1] challenges, since it requires a large number of Poisson components to generate meaningful dynamics, and it uses these additional random components, it cannot be estimated efficiently using the Kalman Filter.

For these reasons, some of the econometric literature focused on developing models that approximate Long Range Dependence within a range of lags of interest. Within discrete time models, the heterogeneous autoregression of LeBaron (2001) and Corsi (2009) reproduces the slow decay of realized volatility by fitting a Heterogeneous Auto-Regressive (HAR) model to its past values, and Buccheri and Corsi (2019) generalizes the model by making its coefficients time varying in the score-

[1]Another criticism often leveled against these models is that unlike some physical systems, in economics and finance there is little evidence and theoretical justification of multifractal behavior, making these models unsuited for structural modeling (see Bouchaud et al. (2000)).

driven framework of Creal et al. (2013) (SHAR) and by accounting for measurement noise with a Kalman filter (SHARK). This approach successfully captures Long Range Dependence, but relies on having meaningful lagged observations to generate long memory. One instance in which this requirement may not be fulfilled is modeling quotes of derivative contracts, such as options, since the number of observed quotes and key characteristics such as moneyness and time to expiry change through time, making it not obvious what to actually use as lagged variables (e.g. the option quote itself through time, or a different option quote with similar characteristics to the current option quote).

This thesis develops a methodology to construct models that attain Long Range Dependence while preserving a finite first-order Markovian state, thus overcoming most of the computational challenges of fractional integration and multiplicative cascades, without requiring lagged observations as in HAR-like models. It constructs dynamics whose autocovariance reproduces a target power-law decay over an arbitrary, pre-specified range of horizons, with the state remaining first-order Markov and of fixed dimension, so that the full history is not required during filtering.

Slow decay in autocovariance of the state is produced by autoregressive components driven by a small set of structural parameters that utilize the approximation procedure of Bochud and Challet (2007). While this approximation procedure was used to replicate long memory in the context of the weights of Neural Networks (Challet and Ragel (2024)), and to create efficient simulations for large covariance matrices (Shu and Nan (2019), Zhang et al. (2021)), this thesis is the first work to apply it to a general time series model with a power law autocovariance in the state equation, allowing it to be used for different modeling scenarios.

The basic model specification requires the autoregressive components to be independent of each other, making it straightforward to implement in a Gaussian State-Space model. Moreover, through the addition of a heuristic approximation, the same structure can be implemented in an observation driven model, thus allowing the use of non-Gaussian features without using simulation-based estimation of parameters.

This approximation resembles the Markovian lifting procedure used in recent mathematical finance literature. For example, Abi Jaber (2019) approximates a Rough-Heston model (that is, a Heston Stochastic Volatility Model with power law (rough) kernel in the variance process) by the sum of geometrically spaced Ornstein-Uhlenbeck processes. However, this procedure has two major differences from Markovian lifting. First, it does not increase the number of parameters depending on how many components are chosen to be used, while Markovian lifting often requires to estimate specific parameters for each of the autoregressive components, which is a convenient advantage when the procedure is applied in a multivariate context, where estimating parameters for increasingly many VAR components may become challenging. Second, this procedure is used to approximate the power law at moderate to large lags, making it useful in situations in which Long Range Dependence needs to be included for various horizons, whereas markovian lifting is often applied for very short lags only, making it suitable in short-term contexts, such as volatility trading or market-making.

The methodology is developed and tested on the Black-Scholes-Merton (Black and Scholes (1973)) Implied Volatility (IV) surface of the S&P 500, the option-implied volatilities quoted across strikes and maturities at each end of day. The surface can be represented by a low-dimensional dynamic factor model with time-varying level and shape as in the Score-Driven and State-Space factor models of Zou et al. (2025), which propose a covariance adjustment to the Score-Driven model to make its density forecast on par with the State-Space model. The approximating construction is applied to such models and compared against a model with true Long Range Dependence, the short-memory specifications of Zou et al. (2025), and a multifractal alternative adapted from the Markov-switching multifractal model of Calvet and Fisher (2004).

The first evaluation is performed through a Monte Carlo study, in which data is generated from a Fractionally Integrated Score-Driven model. When the DGP has long memory, the approximating model is able to capture its dynamics, and outperforms the short-memory models in point forecast up to several months of horizon. Then an empirical forecasting application is conducted on the S&P 500 European IV Option data from OptionsDX.com. The empirical results suggest that while the approximating models' forecasting accuracy is similar to their true long memory counterparts, both specifications have statistically indistinguishable performances to the short-memory models, thus signaling that the curve does not have long memory properties. This result holds for both the whole out-of-sample period using the model confidence set of Hansen et al. (2011) and sequentially through time using the model confidence sequence of Arnold et al. (2026).

The rest of the thesis is organized as follows. Section 2 reviews fractional integration and how it can create computational challenges, and then introduces the Markovian approximation procedure that overcomes these challenges in a general setting. Section 3 defines the models used in the next sections, Section 4 provides evidence of the performance of the approximation procedure in a simulation study, while Section 5 provides the empirical results. Section 6 concludes. Additional empirical and technical results are available in the appendix.

2 Long Range Dependence

Section 2.1 introduces fractional integration as a way to model Long Range Dependence dynamics. After highlighting the limitations of this procedure, Section 2.2 proposes an alternative procedure that generates Long Range Dependence without such limitations.

2.1 Fractional Integration

The traditional approach to generating Long Range Dependence processes is fractional integration. Let $\boldsymbol{y}_t \in \mathbb{R}^{N_t}$ be a (possibly multivariate with time-varying dimension) fractionally integrated process, then $\boldsymbol{y}_t$ depends on a state process $\boldsymbol{\beta}_t \in \mathbb{R}^p$ $p \leq N_t$ that follows a fractional difference equation of the form:

$$(1-L)^{\boldsymbol{d}}\boldsymbol{\beta}_t = \boldsymbol{u}_t, \quad \boldsymbol{\beta}_t = g(\boldsymbol{y}_t;\boldsymbol{\theta}) \tag{1}$$

Here L is the lag operator, $\boldsymbol{d} \in \mathbb{R}^p$, $p \leq N_t$ is the fractional integration parameter, $(1-L)^{\boldsymbol{d}}$ denotes the element-wise exponentiation, $g(\boldsymbol{y};\boldsymbol{\theta}) : \mathbb{R}^{N_t} \to \mathbb{R}^p$ is a measurable function with parameter vector $\boldsymbol{\theta}$ that maps the observed data to a state variable $\boldsymbol{\beta}_t$ and $\boldsymbol{u}_t \in \mathbb{R}^p$ is a process with absolutely summable autocovariances, i.e. the autocovariance function $\boldsymbol{\gamma}_{\boldsymbol{u}}(\tau)$ is such that $\sum_{\tau=-\infty}^{\infty} \|\boldsymbol{\gamma}_{\boldsymbol{u}}(\tau)\| < \infty$ for a matrix norm $\|\cdot\|$.

$\boldsymbol{\beta}_t$ has 'Long Range Dependence' in the sense that under covariance stationarity and $|\boldsymbol{d}_i| < 0.5$, $i = 1,\dots,p$ the autocovariance of the state variable $\boldsymbol{\gamma}_{\boldsymbol{\beta}}(\tau)$ decays hyperbolically:

$$[\boldsymbol{\gamma}_{\boldsymbol{\beta}}(\tau)]_{ii} \sim \boldsymbol{c}_i\tau^{-\boldsymbol{\eta}_i}, \quad \boldsymbol{\eta} = 1 - 2\boldsymbol{d}, \quad \text{as } \tau \to \infty \tag{2}$$

whereas a 'Short-Memory' process's autocovariance decays exponentially. Note that if $\boldsymbol{d}$ were allowed to be in $\mathbb{N}^p$, the process would be a traditional Integrated process of (mixed) order $\boldsymbol{d}$, with the difference that $\boldsymbol{d}$ is not chosen a priori (i.e. a hyperparameter) but estimated using data. Some of the literature denotes as Long Range Dependence processes only those models with $\boldsymbol{d}_i > 0$, since they have positive autocovariance, but here this distinction is not explored, since the methods also apply to the negative cases. Although further generalizations could be considered,[2] Equation (1) nests some of the best-known models in the fractional integration literature. For instance, for scalar y_t and appropriate $\boldsymbol{u}_t$, setting $g(y_t;\boldsymbol{\theta}) = y_t - \mu$ with location parameter μ recovers the ARFIMA$(0,d,0)$ location model of Granger and Joyeux (1980) and Hosking (1981); $g(y_t;\boldsymbol{\theta}) = (y_t - \mu)^2$ recovers the FIGARCH model of Baillie et al. (1996) and $g(y_t;\boldsymbol{\theta}) = (y_t - \mu)^2 - \omega$, $\omega \in \mathbb{R}^+$ is the LMGARCH model of Conrad and Karanasos (2006).

In these standard forms, fractionally integrated models can often be estimated by maximum likelihood as they are standard observation-driven models (as in the classification of Cox (1981)), and often univariate ones. However, the fractional lag operator may cause computational challenges when it needs to be included in more complex data. As a motivating example, consider a general linear Gaussian State-Space model with observations $\boldsymbol{y}_t$ and state $\boldsymbol{\alpha}_t$:

$$\begin{aligned} \boldsymbol{y}_t &= \boldsymbol{h}_t + \boldsymbol{Z}_t\boldsymbol{\alpha}_t + \boldsymbol{\varepsilon}_t, & \boldsymbol{\varepsilon}_t &\sim \mathcal{N}(\boldsymbol{0},\boldsymbol{H}_t) \\ \boldsymbol{\alpha}_{t+1} &= \boldsymbol{m}_t + \boldsymbol{T}_t\boldsymbol{\alpha}_t + \boldsymbol{\xi}_t, & \boldsymbol{\xi}_t &\sim \mathcal{N}(\boldsymbol{0},\boldsymbol{Q}_t) \end{aligned} \tag{3}$$

[2] For example, the literature has proposed parametric and nonparametric methods for fractional co-integration (see Hualde and Robinson (2007) and Johansen and Nielsen (2012))

where $\boldsymbol{\varepsilon}_t$ and $\boldsymbol{\xi}_t$ are independent through time and independent of each other, $\boldsymbol{h}_t, \boldsymbol{m}_t, \boldsymbol{Z}_t, \boldsymbol{T}_t$ are measurable functions. Then the Kalman filter provides the minimum mean square prediction:

$$
\begin{aligned}
\boldsymbol{v}_t &\equiv \boldsymbol{y}_t - \mathbb{E}[\boldsymbol{y}_t|\mathcal{F}_{t-1}] = \boldsymbol{y}_t - \boldsymbol{Z}_t\boldsymbol{a}_t - \boldsymbol{h}_t \\
\boldsymbol{F}_t &\equiv \mathbb{V}\mathrm{ar}[\boldsymbol{y}_t|\mathcal{F}_{t-1}] = \boldsymbol{Z}_t\boldsymbol{P}_t\boldsymbol{Z}_t^\top + \boldsymbol{H}_t \\
\boldsymbol{a}_{t|t} &\equiv \mathbb{E}[\boldsymbol{\alpha}_t|\mathcal{F}_t] = \boldsymbol{a}_t + \boldsymbol{P}_t\boldsymbol{Z}_t^\top\boldsymbol{F}_t^{-1}\boldsymbol{v}_t \\
\boldsymbol{P}_{t|t} &\equiv \mathbb{V}\mathrm{ar}[\boldsymbol{\alpha}_t|\mathcal{F}_t] = \boldsymbol{P}_t - \boldsymbol{P}_t\boldsymbol{Z}_t^\top\boldsymbol{F}_t^{-1}\boldsymbol{Z}_t\boldsymbol{P}_t \\
\boldsymbol{a}_{t+1} &\equiv \mathbb{E}[\boldsymbol{\alpha}_{t+1}|\mathcal{F}_t] = \boldsymbol{T}_t\boldsymbol{a}_{t|t} + \boldsymbol{m}_t \\
\boldsymbol{P}_{t+1} &\equiv \mathbb{V}\mathrm{ar}[\boldsymbol{\alpha}_{t+1}|\mathcal{F}_t] = \boldsymbol{T}_t\boldsymbol{P}_{t|t}\boldsymbol{T}_t^\top + \boldsymbol{Q}_t
\end{aligned} \tag{4}
$$

where $\mathcal{F}_t$ denotes the information set available at time t. The static parameters can be estimated by maximizing the log likelihood:

$$
\mathcal{L}(\boldsymbol{\theta}) = -\frac{1}{2}\sum_{t=1}^{T}(\log|2\pi\boldsymbol{F}_t| + \boldsymbol{v}_t^\top\boldsymbol{F}_t^{-1}\boldsymbol{v}_t) \tag{5}
$$

Now further assume that the state equation $\boldsymbol{\alpha}_t$ is driven by a fractionally integrated component $\boldsymbol{\beta}_t$ as in Equation (1). By expanding the fractional lag operator:

$$
\begin{aligned}
(1-L)^{-\boldsymbol{d}} &= \sum_{\tau=0}^{\infty}\boldsymbol{W}_{FI}(\tau,\boldsymbol{d})L^\tau \\
[\boldsymbol{W}_{FI}(\tau,\boldsymbol{d})]_{i,j} &= \begin{cases} \frac{\Gamma(\tau+\boldsymbol{d}_i)}{\Gamma(\boldsymbol{d}_i)\Gamma(\tau+1)} & i=j \\ 0 & i\neq j \end{cases}
\end{aligned} \tag{6}
$$

where $\Gamma(d)$ denotes the Gamma function and the subscript 'FI' indicates the weights from fractional integration. The dynamics of the fractional component are obtained:

$$
\begin{aligned}
\boldsymbol{\beta}_t &= \sum_{\tau=0}^{\infty}\boldsymbol{W}_{FI}(\tau,\boldsymbol{d})\boldsymbol{u}_{t-\tau} \\
&\approx \sum_{\tau=0}^{t}\boldsymbol{W}_{FI}(\tau,\boldsymbol{d})\boldsymbol{u}_{t-\tau}
\end{aligned} \tag{7}
$$

where the second line is an expanding finite-sample approximation. Clearly, from Equation (7) the initialized state depends on an increasing number of past innovations. To make Equation (7) consistent with the linear Gaussian State-Space, Equation (3) has the form:

$$
\boldsymbol{\alpha}_t = \begin{pmatrix} \boldsymbol{u}_t \\ \boldsymbol{u}_{t-1} \\ \vdots \\ \boldsymbol{u}_1 \end{pmatrix} \in \mathbb{R}^{pt} \quad \boldsymbol{Z}_t = \boldsymbol{X}_t\tilde{\boldsymbol{W}}_t \quad \tilde{\boldsymbol{W}}_t = \begin{pmatrix} \boldsymbol{W}_{FI}(0,\boldsymbol{d}) & \dots & \boldsymbol{W}_{FI}(t-1,\boldsymbol{d}) \end{pmatrix} \tag{8}
$$

$$
\boldsymbol{T}_t = \begin{pmatrix} \boldsymbol{0}_{p\times pt} \\ \boldsymbol{I}_{pt} \end{pmatrix} \quad \boldsymbol{\xi}_t = \begin{pmatrix} \boldsymbol{u}_{t+1} \\ \boldsymbol{0}_{pt} \end{pmatrix} \quad \boldsymbol{Q}_t = \begin{pmatrix} \boldsymbol{Q}_u & \boldsymbol{0}_{p\times pt} \\ \boldsymbol{0}_{pt\times p} & \boldsymbol{0}_{pt\times pt} \end{pmatrix} \quad \boldsymbol{m}_t = \boldsymbol{0}_{p(t+1)} \tag{9}
$$

where $\boldsymbol{Q}_u \in \mathbb{R}^{p\times p}$ is the covariance of $\boldsymbol{u}_t$, $\boldsymbol{X}_t \in \mathbb{R}^{N_t\times p}$ is an $\mathcal{F}_t$-measurable matrix that determines the linear effect of $\boldsymbol{\beta}_t = \tilde{\boldsymbol{W}}_t\boldsymbol{\alpha}_t$ to $\boldsymbol{y}_t$ (assuming $\boldsymbol{u}_0 = \boldsymbol{0}_p$). The process has a state space representation that increases in size at each t, and the limiting process has an infinite dimensional state. For a large enough sample size, this requires extensive computer memory costs, and the Kalman filter requires computations of very large matrix objects.

For this reason, the most common approach to model $\boldsymbol{\beta}_t$ is to introduce a static truncation, rather than expanding the state equation with the number of observations. However, this approximation only reaches the asymptotic power-law autocovariance if the truncation includes a large number of lagged components, thus possibly still retaining the disadvantages of the exact model, with the additional disadvantage of treating the current state as independent of the innovations after the truncation lag. Therefore, cases such as State-Space models, even in the linear Gaussian case, may be infeasible to implement with fractionally integrated components due to the large computer memory requirements they face. Similar problems are encountered in Random Multiplicative Cascade models, which implement the power law decay through hierarchical Hidden Markov Models (one example of such models is provided in Section 3.3), as their forward filter requires storing very large transition matrices to effectively capture power law decay, and their estimation in a State-Space setting requires simulation-based inference.

2.2 Markovian Approximation Procedure

To approximate the power law decay without the computational challenges of fractionally integrated processes, a similar approach to Markovian lifting from the mathematical finance literature is used. To understand this procedure, note that Long Range Dependence processes often admit an infinite Markovian representation. To see this, consider a process with autocorrelation decaying as a power law of time $\tau^{-\eta}$, $\eta \in (0, 2)$. From the definition of the Gamma function:

$$\Gamma(\eta) = \int_0^\infty u^{\eta-1}\exp(-u)\,du \tag{10}$$

let $u = \tau\lambda$, so that $du/d\lambda = \tau \longleftrightarrow du = \tau\,d\lambda$ and:

$$\begin{aligned}\Gamma(\eta) &= \int_0^\infty (\lambda\tau)^{\eta-1}\exp(-\tau\lambda)\tau\,d\lambda \\ &= \int_0^\infty \lambda^{\eta-1}\tau^{\eta-1}\exp(-\tau\lambda)\tau\,d\lambda \\ &= \tau^\eta \int_0^\infty \lambda^{\eta-1}\exp(-\tau\lambda)\,d\lambda\end{aligned} \tag{11}$$

multiply both sides of Equation (11) by $\tau^{-\eta}/\Gamma(\eta)$ to find the following representation:

$$\tau^{-\eta} = \frac{1}{\Gamma(\eta)}\int_0^\infty \lambda^{\eta-1}e^{-\tau\lambda}\,d\lambda \tag{12}$$

from Equation (12), it becomes clear that the power law autocorrelation is composed of an infinite sum of exponential components. Then intuitively, one may try to find a finite sum that approximates this integral, with components of the form:

$$\tau^{-\eta} \approx \frac{1}{\Gamma(\eta)} \sum_{i=0}^{K} \lambda_i^{\eta-1} e^{-\tau\lambda_i} \Delta_i \tag{13}$$

where Δ_i are quadrature spacings. If such finite sum exists, then the state with Long-Range dependence can be effectively approximated by a state given by the scaled sum of a finite number K of Markovian processes with exponential decay in autocorrelation $\lambda_i^{\eta-1} e^{-\tau\lambda_i}$.

One challenge is that the approximation must be precise enough not only for a single lag τ, but rather for a range of lags of interest. Moreover, to make the approximation precise in certain lag ranges, it may be required to use a large number of components K, which may introduce a curse of dimensionality into the estimation.

The procedure of Bochud and Challet (2007) is designed to find an approximation of the form of Equation (13) that overcomes these challenges. Let $w_k = \Delta_k \lambda_k^{\eta-1}/\Gamma(\eta)$ and $g(\tau) = \sum_{k=0}^{K} w_k e^{-\tau\lambda_k}$ the approximating autocovariance. The following is a kernel with an explicit set of w_k and λ_k that approximates the power law autocorrelation for a range of lags $\tau_0, \ldots, \tau_K$ of interest. To understand the procedure, first the choice of the lags of interest is qualified, from which the choice of w_k and λ_k follows.

For the choice of the target lags, consider the relative approximation error $|\tau^{-\eta} - g(\tau)|/\tau^{-\eta}$ on two consecutive intervals $[\tau_k, \tau_{k+1}]$ and $[\tau_{k+1}, \tau_{k+2}]$ with $\tau_{k+1} = \phi\tau_k$ and $\tau_{k+2} = \phi\tau_{k+1}$ for some $\phi > 1$. On the second interval the target equals $\phi^{-\eta}$ times the target on the first, since $(\phi\tau)^{-\eta} = \phi^{-\eta}\tau^{-\eta}$ for all τ. Then if $g(\tau)$ achieves relative error ϵ on $[\tau_k, \tau_{k+1}]$, meaning $|\tau^{-\eta} - g(\tau)|/\tau^{-\eta} \leq \epsilon$ in this interval, then scaling g by $\phi^{-\eta}$ achieves the same relative error ϵ on $[\tau_{k+1}, \tau_{k+2}]$, since numerator and denominator both scale by $\phi^{-\eta}$. Each interval of proportional width ϕ therefore presents the same approximation problem.

For this reason, the specific lag values at which $g(\tau)$ is constrained to match $\tau^{-\eta}$ are:

$$\tau_k = \phi^k, \quad k = 0, \ldots, K \tag{14}$$

Clearly, different ϕ generate different approximations. In Section 3.4 $\phi > 1$ is estimated by Maximum Likelihood, however, if a certain region of lags is of particular interest, one could also treat it as a hyperparameter. Moving to the choice of $\{\lambda_k\}_{k=0}^{K}$, this specification aims to have one component $w_k e^{-\lambda_k\tau}$ per interval sufficient to achieve a uniform relative error ϵ across all intervals, so that $K+1$ components would cover the entire target range with one component per interval. In particular, k-th component $w_k e^{-\lambda_k\tau}$ is assigned to approximate $\tau^{-\eta}$ at its collocation point $\tau_k = \phi^k$, treating the contributions of all other terms $w_j e^{-\lambda_j\tau}$ for $j \neq k$ as negligible at τ_k (the accuracy of this assumption is quantified below). This gives the following two conditions:

$$w_k e^{-\lambda_k \tau_k} = \phi^{-k\eta} \tag{15}$$

$$\frac{d\ln(w_k e^{-\lambda_k \tau})}{d\ln\tau} = \frac{d\ln(\tau^{-\eta})}{d\ln\tau} \tag{16}$$

Expanding the left-hand side of Equation (16):

$$\frac{d\ln(w_k e^{-\lambda_k \tau})}{d\ln\tau} = \frac{\tau}{w_k e^{-\lambda_k \tau}} \cdot (-\lambda_k w_k e^{-\lambda_k \tau}) = -\lambda_k \tau \tag{17}$$

and setting Equation (17) equal to $-\eta$ at $\tau = \tau_k = \phi^k$:

$$-\lambda_k \phi^k = -\eta \implies \lambda_k = \eta\phi^{-k} \tag{18}$$

With $\lambda_k = \eta\phi^{-k}$ established from Equation (18), the contribution of the j-th term $w_j e^{-\lambda_j \tau}$ to $g(\tau)$ at collocation point $\tau_k = \phi^k$ is, exactly:

$$w_j e^{-\lambda_j \tau_k} = w_j \exp(-\eta\phi^{k-j}) \tag{19}$$

For $j < k$: the exponent is $-\eta\phi^{k-j}$ with $k - j \geq 1$, giving $e^{-\eta\phi^{k-j}} \leq e^{-\eta\phi}$, since the j-th term has $1/\lambda_j = \phi^j/\eta \ll \tau_k$ and has decayed to a value at most $e^{-\eta\phi}$ by lag τ_k. For $j > k$: expanding $e^{-\eta\phi^{-(j-k)}} = 1 - \eta\phi^{-(j-k)} + O(\phi^{-2(j-k)})$ shows that the j-th term contributes exactly $w_j(1 - \eta\phi^{-(j-k)} + O(\phi^{-2(j-k)}))$ to $g(\tau_k)$, since the j-th term has $1/\lambda_j = \phi^j/\eta \gg \tau_k$ and has not yet reached its transition point at τ_k. The total leakage at τ_k, meaning the exact combined contribution of all $j > k$ terms to $g(\tau_k)$, is:

$$\sum_{j=k+1}^{K} w_j \exp(-\eta\phi^{-(j-k)}) = \sum_{j=k+1}^{K} w_j(1 - \eta\phi^{-(j-k)} + O(\phi^{-2(j-k)})) \tag{20}$$

Then a sensible choice for the weights is $w_k = c_k\phi^{-k\eta}$, where c_k is a correction coefficient that accounts for the leakage.

To incorporate the leakage in the c_k terms, note that from the condition $g(\tau_k) = \phi^{-k\eta}$ for $k = 0, \ldots, K$, substituting $w_j = c_j\phi^{-j\eta}$ and $\lambda_j = \eta\phi^{-j}$, becomes exactly:

$$\sum_{j=0}^{K} c_j \phi^{-j\eta} e^{-\eta\phi^{k-j}} = \phi^{-k\eta}, \quad k = 0, \ldots, K \tag{21}$$

Dividing both sides of Equation (21) by $\phi^{-k\eta}$ and separating the sum by whether $j < k$, $j = k$, or $j > k$:

$$c_k e^{-\eta} + \sum_{j=0}^{k-1} c_j \phi^{(k-j)\eta} e^{-\eta\phi^{k-j}} + \sum_{j=k+1}^{K} c_j \phi^{-(j-k)\eta} e^{-\eta\phi^{-(j-k)}} = 1 \tag{22}$$

The sum over $j < k$ contains terms each bounded above by $c_j\phi^{(k-j)\eta}e^{-\eta\phi^{k-j}} \leq c_j\phi^{(k-j)\eta}e^{-\eta\phi}$, an approximation error of order $e^{-\eta\phi}$ relative to $c_k e^{-\eta}$, decreasing rapidly as ϕ increases. Dropping this sum as an approximation with relative error of order $e^{-\eta\phi}$ and isolating c_k:

$$c_k e^{-\eta} = 1 - \sum_{j=k+1}^{K} c_j \phi^{-(j-k)\eta} e^{-\eta\phi^{-(j-k)}} \tag{23}$$

Multiplying both sides by e^{η} and applying the exact identity $e^{\eta} \cdot \phi^{-(j-k)\eta} \cdot e^{-\eta\phi^{-(j-k)}} = \phi^{-(j-k)\eta} \cdot e^{\eta(1-\phi^{-(j-k)})}$, which follows from $e^{\eta} \cdot e^{-\eta\phi^{-(j-k)}} = e^{\eta(1-\phi^{-(j-k)})}$:

$$c_k = \left(e^{\eta} - \sum_{j=k+1}^{K} c_j \phi^{-(j-k)\eta} \cdot e^{\eta(1-\phi^{-(j-k)})} \right) \tag{24}$$

The factor e^{η} is a positive constant independent of k that multiplies every c_k identically and therefore does not impact the solution with respect to c_k. Then the equivalent system to solve is:

$$c_k = 1 - \sum_{j=k+1}^{K} c_j \cdot \phi^{-(j-k)\eta} \cdot e^{\eta(1-\phi^{-(j-k)})}, \quad k = 0, \dots, K \tag{25}$$

This is a triangular system of $K+1$ equations: c_k depends only on $c_{k+1}, \dots, c_K$, so the unknowns are solved sequentially from $k = K$, where the sum is empty and $c_K = 1$, through $c_{K-1}, c_{K-2}, \dots, c_0$, each using only previously computed values. Re-indexing Equation (25) with $k \to K-k$ and $j \to K-i$:

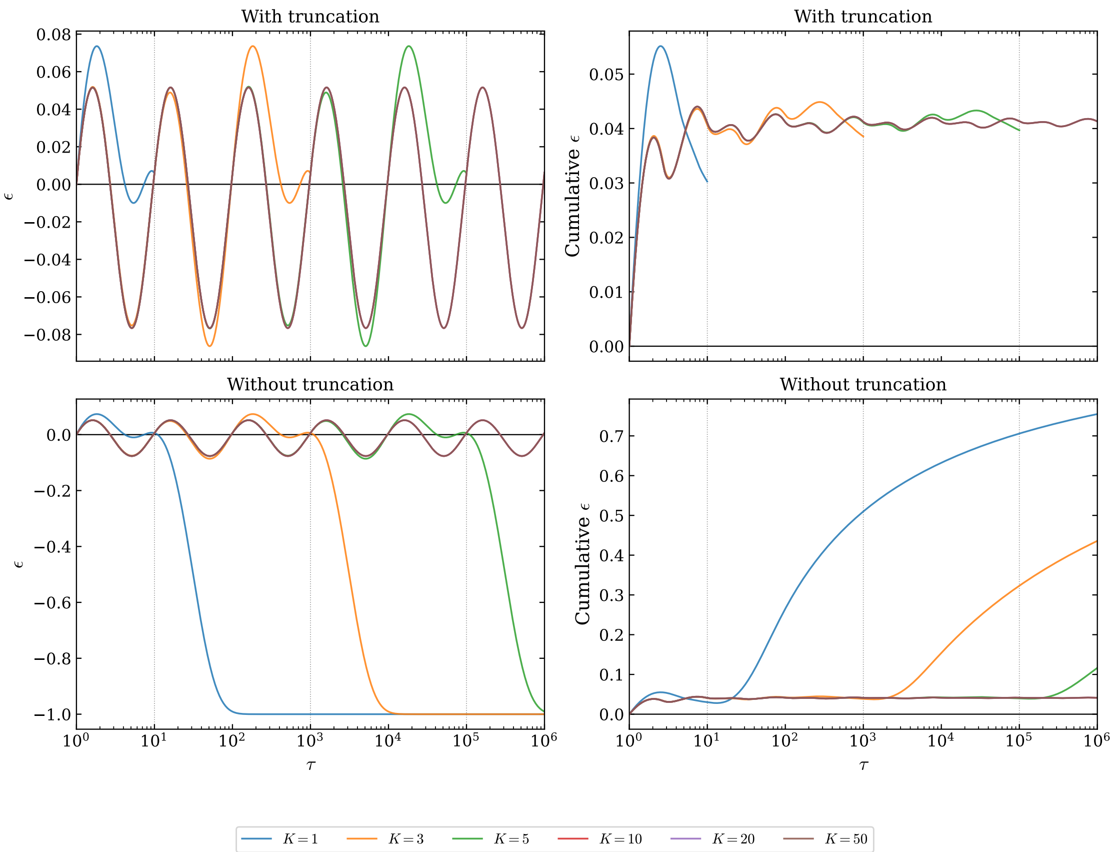
$$c_{K-k} = 1 - \sum_{i=0}^{k-1} c_{K-i} \cdot \phi^{-\eta(k-i)} \cdot e^{\eta(1-\phi^{-(k-i)})}, \quad c_K = 1 \tag{26}$$

Each term $c_{K-i}\phi^{-\eta(k-i)}e^{\eta(1-\phi^{-(k-i)})}$ is, exactly, the leakage contribution of the $(K-i)$-th term $w_{K-i}e^{-\lambda_{K-i}\tau}$ at matching point τ_{K-k}: $\phi^{-\eta(k-i)} = \tau_{K-i}^{-\eta}/\tau_{K-k}^{-\eta}$ is the ratio of the base weight $\phi^{-(K-i)\eta}$ of the $(K-i)$-th term to the base weight $\phi^{-(K-k)\eta}$ of the $(K-k)$-th term, and $e^{\eta(1-\phi^{-(k-i)})} = e^{-\eta\phi^{-(k-i)}}/e^{-\eta}$ is the value of the $(K-i)$-th exponential $e^{-\lambda_{K-i}\tau_{K-k}}$ normalized by the diagonal factor $e^{-\eta}$.

The correction coefficient c_{K-k} reduces the base weight $\phi^{-(K-k)\eta}$ of the $(K-k)$-th term by exactly this aggregate leakage, so that the matching condition at τ_{K-k} is satisfied to the approximation order established above.

As an illustrative example, Figure 1 shows the relative error of the approximation over a range of lags for a power law decay. As expected, for $\tau_i = \phi^i$, $i \in (0, K)$ the approximation procedure matches exactly the power law. The approximation is most precise over the range $[\tau_0, \tau_K] = [1, \phi^K]$: for $\tau < 1$ all components have not yet reached their transition point $1/\lambda_k$ and $g(\tau)$ behaves as a constant rather than a power law; for $\tau > \phi^K$ all components have decayed past their transition points and $g(\tau) \approx 0$ while $\tau^{-\eta} > 0$. Between the lags of interest the error is of order $e^{-\eta\phi}$ due to the dropped $j < k$ terms. Moreover, 2 approximations with the same ϕ and different numbers of components, say $K_1 > K_2$, will produce identical approximation quality

Figure 1: Relative Approximation Error against true power law.



Note: The true power law uses $\eta = 0.7$ and the approximations use $\phi = 10$. Gray vertical lines delimit the boundary points of the approximation range. First row shows the absolute relative error and the cumulative relative error truncated to the boundary region, second row does not clip.

in the range $[1, \phi^{K_2}]$, regardless of how much larger K_1 is, which implies that a larger number of components should be used only if reproducing memory over the more distant past.

Then, to implement the procedure, consider a process with a state α_t with power law autocovariance of the form $\sigma_\xi^2 \tau^{-\eta}$, $\tau > 0$, $\eta \in (0, 2)$, $\sigma_\xi^2 > 0$. Generally, the state dimension increases in t. Instead, if only certain lags are of interest, the state's autocorrelation can be effectively approximated by a model of the form:

$$\begin{aligned}
\alpha_t &\approx \sum_{i=0}^{K} w_i b_{i,t} \\
b_{i,t+1} &= \rho_i b_{i,t} + \sqrt{\frac{1-\rho_i^2}{w_i}}\xi_{i,t} \\
\rho_i &= \exp\left(-\eta\phi^{-i}\right) \\
w_i &= c_i \phi^{-i\eta} e^{\eta} \\
c_K &= 1 \\
c_{K-i} &= 1 - \sum_{j=0}^{i-1} c_{K-j}\phi^{-\eta(i-j)} \exp(\eta(1-\phi^{-(i-j)}))
\end{aligned} \tag{27}$$

where $\xi_{i,t}$ are independent, identically distributed shocks with zero mean and common variance σ_ξ^2 across t and i. Under this process, the autocovariance function of the state:

$$\begin{aligned}
\mathbb{C}\mathrm{o}\mathbb{V}(\alpha_t;\alpha_{t-\tau}) &= \sum_{i=0}^{K}\sum_{j=0}^{K} w_i w_j \mathbb{C}\mathrm{o}\mathbb{V}(b_{i,t};b_{j,t-\tau}) \\
\mathbb{C}\mathrm{o}\mathbb{V}(b_{i,t};b_{j,t-\tau}) &= \mathbb{C}\mathrm{o}\mathbb{V}\left(\rho_i b_{i,t-1} + \sqrt{\frac{1-\rho_i^2}{w_i}}\xi_{i,t};b_{j,t-\tau}\right) \\
&= \rho_i \mathbb{C}\mathrm{o}\mathbb{V}(b_{i,t-1};b_{j,t-\tau}) + \underbrace{\sqrt{\frac{1-\rho_i^2}{w_i}}\mathbb{C}\mathrm{o}\mathbb{V}(\xi_{i,t};b_{j,t-\tau})}_{=0} \\
&= \rho_i^\tau \underbrace{\mathbb{C}\mathrm{o}\mathbb{V}(b_{i,t-\tau};b_{j,t-\tau})}_{0 \text{ if } i\neq j,\, \mathbb{V}\mathrm{ar}(b_{i,t}) \text{ if } i=j} \\
\mathbb{V}\mathrm{ar}(b_{i,t}) &= \rho_i^2 \mathbb{V}\mathrm{ar}(b_{i,t}) + \frac{1-\rho_i^2}{w_i}\sigma_\xi^2 \\
&= \sigma_\xi^2 / w_i \\
\Rightarrow \mathbb{C}\mathrm{o}\mathbb{V}(\alpha_t;\alpha_{t-\tau}) &= \sum_{i=0}^{K} w_i^2 \rho_i^\tau \frac{\sigma_\xi^2}{w_i} \\
&= \sigma_\xi^2 \sum_{i=0}^{K} w_i \exp(-\eta\phi^{-i}\tau)
\end{aligned} \tag{28}$$

which recovers the approximation formula for the procedure. Note that if σ_ξ^2 is not a free scalar, to make a comparison between the DGP and the approximating model it is convenient to restrict this term. For instance, assume the state α_t follows an ARFIMA$(0,d,0)$ process, then its (asymptotic) autocovariance is:

$$\sigma_{FI}^2 \frac{\Gamma(1-2d)}{\Gamma(d)\Gamma(1-d)}\tau^{2d-1} \tag{29}$$

in this case, estimating a free σ^2_ξ can still capture the overall dynamics of the process, but to directly interpret σ^2_ξ as an estimate for σ^2_{FI} the variance of the innovations should be scaled by $\Gamma(\eta)/\Gamma((1-\eta)/2)\Gamma((1+\eta)/2)$ since $\eta = 1-2d$.

More generally, if the power law autocovariance is scaled by a term D independent of τ, the $b_{i,t}$ terms can be adjusted as:

$$b_{i,t+1} = \rho_i b_{i,t} + \sqrt{D\frac{1-\rho_i^2}{w_i}}\xi_{i,t} \tag{30}$$

so that the approximating covariance adjusts to:

$$\mathbb{C}\mathrm{o}\mathbb{V}(\alpha_t;\alpha_{t-\tau}) = D\sum_{i=0}^{K} w_i \exp(-\eta\phi^{-i}\tau) \tag{31}$$

which again recovers the approximation multiplied by the additional term D.
Note that the approximating autocovariance of Equation (28) is recovered due to the independence of the noise terms $\xi_{i,t}$. Instead, if the state is observation driven, this does not hold. Suppose that each $b_{i,t}$ is instead Score-Driven:

$$b_{i,t+1} = \rho_i b_{i,t} + \sqrt{\frac{1-\rho_i^2}{w_i}} s^{(b)}_{i,t} \tag{32}$$

where $s^{(b)}_{i,t}$ denotes the scaled-score of $b_{i,t+1}$ with respect to the log-density of the model. Here the approximating components are driven by $s^{(b)}_{i,t} = w_i s^{(\alpha)}_t$:[3]

$$b_{i,t+1} = \rho_i b_{i,t} + \sqrt{(1-\rho_i^2)w_i} s^{(\alpha)}_t \tag{33}$$

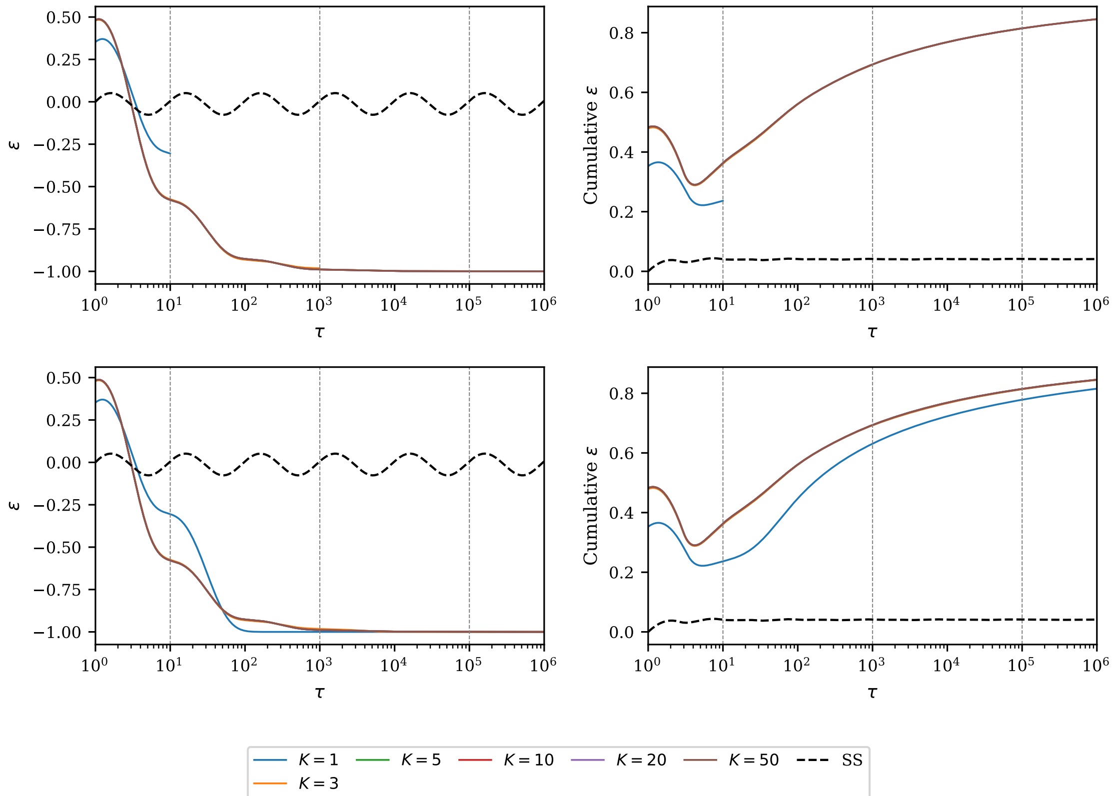
Each of the components is innovated through the common scaled score component $s^{(\alpha)}_t$ and state's autocovariance is:

$$\begin{aligned}
\mathbb{C}\mathrm{o}\mathbb{V}(\alpha_t;\alpha_{t-\tau}) &= \sum_{i=0}^{K}\sum_{j=0}^{K} w_i w_j \rho_i^\tau \mathbb{C}\mathrm{o}\mathbb{V}(b_{i,t-\tau}; b_{j,t-\tau}) \\
\mathbb{C}\mathrm{o}\mathbb{V}(b_{i,t-\tau}; b_{j,t-\tau}) &= \frac{\sqrt{(1-\rho_i^2)w_i}\sqrt{(1-\rho_j^2)w_j}\,\mathbb{V}\mathrm{ar}\left(s^{(\alpha)}_t\right)}{1-\rho_i\rho_j} \\
&\equiv \frac{\sqrt{(1-\rho_i^2)w_i}\sqrt{(1-\rho_j^2)w_j}\,\sigma_s^2}{1-\rho_i\rho_j} \\
\Rightarrow \mathbb{C}\mathrm{o}\mathbb{V}(\alpha_t;\alpha_{t-\tau}) &= \sigma_s^2 \sum_{i=0}^{K}\sum_{j=0}^{K} \frac{w_i^{3/2} w_j^{3/2}\sqrt{(1-\rho_i^2)(1-\rho_j^2)}}{1-\rho_i\rho_j}\rho_i^\tau
\end{aligned} \tag{34}$$

which is misaligned from the approximation procedure. Figure 2 shows how the

[3] An alternative would be to define $s^{(b)}_{i,t} = w_i^{-1} s^{(\alpha)}_t$, which also corresponds to the scaled score with respect to $b_{i,t}$, however this increases numerical instability in the model, and is found to be similar to the chosen specification once the κ_i adjustment of Equation (38) is applied.

Figure 2: Relative Approximation error against true power law in Score-Driven setting



Note: The true power law uses $\eta = 0.7$ and the approximations use $\phi = 10$ and $\sigma_s^2 = 1$. Gray vertical lines delimit the boundary points of the approximation range. Black, dashed line refers to the approximation in the State-Space case and $K = 50$. First row shows the absolute relative error and the cumulative relative error truncated to the boundary region, second row does not clip.

approximation procedure behaves under this specification. Unlike Figure 1, the error is not bounded between lags $\tau_i = \phi^i, i = 1, \ldots, K$ and adding additional components does not improve precision, making the performance much worse than the State-Space approximation.

Note that in the final line of Equation (34) the lag τ enters the autocovariance only through the ρ_i^τ term and not on any j terms, making it possible to rewrite it as:

$$\begin{aligned} \mathbb{C}\mathrm{o}\mathbb{V}(\alpha_t; \alpha_{t-\tau}) &= \sigma_s^2 \sum_{i=0}^{K} \tilde{w}_i \exp(-\eta\phi^{-i}\tau) \\ \tilde{w}_i &= w_i^{3/2}\sqrt{(1-\rho_i^2)} \sum_{j=0}^{K} \frac{w_j^{3/2}\sqrt{(1-\rho_j^2)}}{1-\rho_i\rho_j} \end{aligned} \tag{35}$$

which resembles the correct approximation, although it uses distorted weights $\tilde{w}_i$.

Then intuitively, to make the approximation work as intended, one may multiply the score in Equation (32) by a term κ_i such that the modified dynamics result in $\tilde{w}_i = w_i$, which requires solving the following system for κ_i:

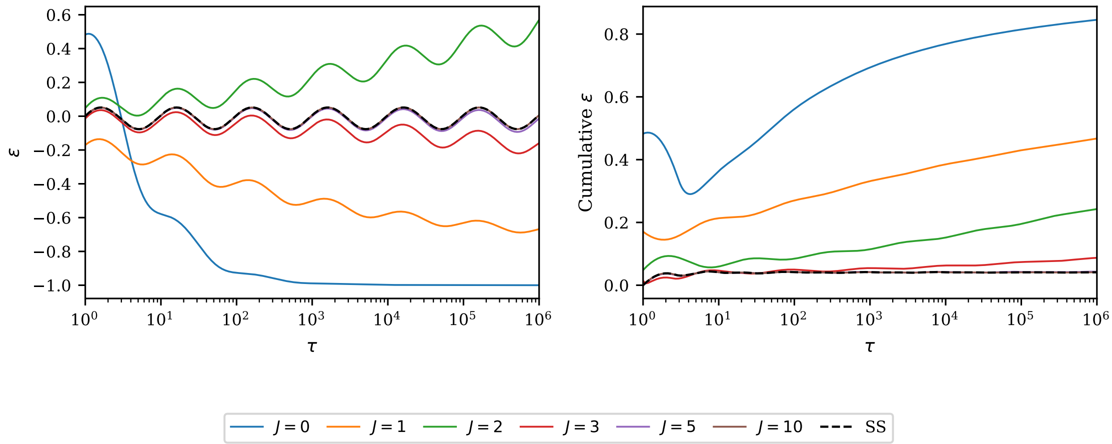
$$
\begin{aligned}
w_i &= \kappa_i w_i^{3/2} \sqrt{(1-\rho_i^2)} \sum_{j=0}^{K} \frac{\kappa_j w_j^{3/2} \sqrt{(1-\rho_j^2)}}{1-\rho_i\rho_j} \\
&= \frac{\kappa_i^2 w_i^3 (1-\rho_i^2)}{1-\rho_i^2} + \kappa_i w_i^{3/2} \sqrt{(1-\rho_i^2)} \sum_{j\neq i}^{K} \frac{\kappa_j w_j^{3/2} \sqrt{(1-\rho_j^2)}}{1-\rho_i\rho_j} \\
&= \kappa_i^2 w_i^3 + \kappa_i w_i^{3/2} \sqrt{(1-\rho_i^2)} \sum_{j\neq i}^{K} \frac{\kappa_j w_j^{3/2} \sqrt{(1-\rho_j^2)}}{1-\rho_i\rho_j} \\
&= \mu_i^2 w_i + \mu_i w_i^{1/2} \sqrt{(1-\rho_i^2)} \sum_{j\neq i}^{K} \frac{\mu_j w_j^{1/2} \sqrt{(1-\rho_j^2)}}{1-\rho_i\rho_j}, \quad \mu_i = w_i \kappa_i \\
\rightarrow 1 &= \mu_i^2 + \mu_i w_i^{-1/2} \sqrt{(1-\rho_i^2)} \sum_{j\neq i}^{K} \frac{\mu_j w_j^{1/2} \sqrt{(1-\rho_j^2)}}{1-\rho_i\rho_j} \\
&= \mu_i^2 + \mu_i \sum_{j\neq i}^{K} L_{i,j}\mu_j, \quad L_{i,j} = \sqrt{\frac{w_j}{w_i}} \frac{\sqrt{(1-\rho_i^2)(1-\rho_j^2)}}{1-\rho_i\rho_j}
\end{aligned}
\tag{36}
$$

the last line of Equation (36) is a system of quadratic equations in μ_i which is challenging to solve analytically. Instead, here the following approximation is proposed. For each component, starting from $i = 0$, the matching condition is enforced by solving the quadratic equation $\mu_i^2 + \mu_i \sum_{j\neq i}^{K} L_{i,j}\mu_j - 1 = 0$ for μ_i by treating the terms $\mu_j\, j < i$ as fixed and ignoring the terms for $j > i$. Intuitively, this approximation works because the $L_{i,j}\, j < i$ terms are smaller by construction, thus generating less impact to the approximation error. Let $z_i = \sum_{j<i}^{K} L_{i,j}\mu_j$, then μ_i is found iteratively by:

$$
\begin{aligned}
\mu_0 &= 1 \\
\mu_i &= \frac{-z_i + \sqrt{z_i^2 + 4}}{2} \\
\kappa_i &= \frac{-z_i + \sqrt{z_i^2 + 4}}{2w_i}
\end{aligned}
\tag{37}
$$

which comes from solving the quadratic equations of the system and substituting κ_i from the definition of μ_i. Note that although 2 roots exist in general, the second root $0.5\left(-z_i - \sqrt{z_i^2+4}\right) < 0$ is always negative, making the condition $\tilde{w}_i = w_i$ impossible to match, since $w_i > 0$. Then, to reduce the approximation error, after computing the terms μ_i from Equation (37), the procedure can be repeated, this time by no longer ignoring the terms μ_j for $j > i$, since an intermediate estimate of them is computed. This procedure can then be repeated until the μ_i terms converge.

Figure 3: Relative Approximation error against true power law for iterative procedure



Note: The true power law uses $\eta = 0.7$ and the approximations use $\phi = 10$, $\sigma_s^2 = 1$ and $K = 50$. J is the number of iterations used to find $\kappa_0, \ldots, \kappa_K$, with $J = 0$ being the non-adjusted Error as in Figure 2

Let $\boldsymbol{\kappa}^{(J)}$ be the vector of adjustment factors computed at iteration J, so that the complete algorithm for the Score-Driven set up is:

$$
\begin{aligned}
\mu_0^{(0)} &= 1 \\
z_i^{(0)} &= \sum_{j<i}^{K} L_{i,j}\, \mu_j^{(0)} \\
\mu_i^{(0)} &= \frac{-z_i^{(0)} + \sqrt{(z_i^{(0)})^2 + 4}}{2} \\
\text{For } J = 1, 2, 3, \ldots \text{ until } \|\boldsymbol{\mu}^{(J)} - \boldsymbol{\mu}^{(J-1)}\| < \epsilon : \\
z_i^{(J)} &= \sum_{j \neq i}^{K} L_{i,j}\, \mu_j^{(J-1)} \\
\mu_i^{(J)} &= \frac{-z_i^{(J)} + \sqrt{(z_i^{(J)})^2 + 4}}{2} \\
\kappa_i &= \frac{\mu_i^{(J)}}{w_i} \\
\implies b_{i,t+1} &= \rho_i b_{i,t} + \kappa_i \sqrt{(1 - \rho_i^2) w_i}\, s_t^{(\alpha)} \\
&= \rho_i b_{i,t} + \mu_i^{(J)} \sqrt{\frac{1 - \rho_i^2}{w_i}}\, s_t^{(\alpha)}
\end{aligned}
\tag{38}
$$

for a tolerance level ϵ and a norm $\| \cdot \|$. Figure 3 shows the relative approximation error of the adjusted procedure against the non-adjusted approximation error in the

Score-Driven and the State-Space autocovariance. Without the iterative procedure of Equation (38) the Score-Driven model's autocovariance is not able to replicate the power law, and the relative error makes it inconsistent. By iterating this process a few times, the set of coefficients κ_i is able to replicate the approximation of the State-Space model. Although a formal proof of convergence is not presented, by testing convergence over a grid of η, ϕ one finds that convergence happens uniformly for $\phi \geq 7.5$ for all $\eta \in (0, 2)$.

Note that this iterative procedure is done once per likelihood guess and converges rapidly, making this model much more convenient than the fractionally integrated procedure, in which all past observations need to be taken into account at each filtering step.

3 Modeling framework

At the end of each trading day, a time-varying number N_t of option quotes are observed. Then, by inverting the Black-Scholes-Merton option pricing formula, the IV can be obtained for each observed strike price and moneyness level. The implied volatility surface is modeled as dynamic factor models of the form:

$$\log \boldsymbol{IV}_t = \boldsymbol{M}_t \boldsymbol{\beta}_t + \boldsymbol{\varepsilon}_t, \quad \boldsymbol{\varepsilon}_t \sim p_\nu \left(\mathbf{0}, \boldsymbol{H}_t; \nu \right), \tag{39}$$

where $\boldsymbol{M}_t$ is the $N_t \times p$ matrix of exogenous factors; $\boldsymbol{\beta}_t$ is the p-dimensional vector of time-varying factor loadings, p_ν is a multivariate distribution with shape parameter ν and $\boldsymbol{H}_t$ is the noise covariance matrix.

$\boldsymbol{\beta}_t$ varies in time using a recursion of the form:

$$\boldsymbol{\beta}_{t+1} = (\boldsymbol{I}_p - \boldsymbol{B})\bar{\boldsymbol{\beta}} + \boldsymbol{B}\boldsymbol{\beta}_t + \boldsymbol{\xi}_t \tag{40}$$

where $\boldsymbol{B} \in \mathbb{R}^{p \times p}$ is a parameter diagonal matrix with all eigenvalues inside the unit circle, $\bar{\boldsymbol{\beta}} \in \mathbb{R}^p$ is a parameter vector, and $\boldsymbol{\xi}_t \in \mathbb{R}^p$ represents a generic updating mechanism.

Each of the following subsections defines a different specification of $\boldsymbol{\xi}_t$. Subsection 3.1 presents the short memory State-Space and adjusted Score-Driven models of Zou et al. (2025); the Long Range Dependence, non-Markovian adjusted score driven model is presented in Subsection 3.2; Subsection 3.3 presents an alternative Long Range Dependence model with Markovian structure. This specification is adapted from the model of Calvet and Fisher (2004), and has 'Multifractal' properties. Finally, Subsection 3.4 presents the Markovian Long Range Dependence models.

3.1 Short-Memory models

Depending on the choice of $\boldsymbol{\xi}_t$ either a State-Space model or a Score-Driven model is recovered. If $\{(\boldsymbol{\varepsilon}_t^\top, \boldsymbol{\xi}_t^\top\}_{t \in \mathbb{Z}}$ is an independently and identically distributed (iid) sequence of innovations with mutually independent Gaussian components $\boldsymbol{\varepsilon}_t$ and $\boldsymbol{\xi}_t$, then Equations (39)-(40) are a standard linear Gaussian State-Space set-up and the shape parameter ν can be omitted. The Kalman filter of Equation (4) can be implemented as:

$$
\begin{aligned}
\boldsymbol{y}_t &= \log \boldsymbol{IV}_t, & \boldsymbol{h}_t &= 0, & \boldsymbol{Z}_t &= \boldsymbol{M}_t \\
\boldsymbol{\alpha}_t &= \boldsymbol{\beta}_t, & \boldsymbol{m}_t &= (\boldsymbol{I}_p - \boldsymbol{B})\bar{\boldsymbol{\beta}}, & \boldsymbol{T}_t &= \boldsymbol{B}, \\
\boldsymbol{H}_t &= \sigma^2 \boldsymbol{I}_{N_t}, & \sigma^2 &\in \mathbb{R}^+, & \boldsymbol{Q}_t &= \mathrm{diag}(q_1, \ldots, q_p) \in \mathbb{R}^{p+}
\end{aligned}
\tag{41}
$$

Then the static parameter vector $\boldsymbol{\theta} = \left(\sigma^2, \mathrm{diag}(\boldsymbol{B})^\top, \mathrm{diag}(\boldsymbol{Q_t})^\top, \bar{\boldsymbol{\beta}}^\top\right)^\top$ can be estimated by maximizing the likelihood function of Equation (5).

For non-Gaussian $\boldsymbol{\varepsilon}_t$ the Kalman Filter still provides the *linear* minimum mean squared error, while sequential Monte Carlo methods are required to approach the 'true' minimum. Such techniques are typically more challenging and computationally intensive.

Instead, if $\boldsymbol{\xi}_t$ is chosen as the derivative with respect to $\boldsymbol{\beta}_t$ of the log predictive density of $\log \boldsymbol{IV}_t$ given $\boldsymbol{\beta}_t$, the model is within the Score-Driven framework of Creal et al. (2013). Then the parameter vector $\boldsymbol{\theta}$ can be estimated by standard Maximum likelihood even if $\boldsymbol{\varepsilon}_t$ is not Gaussian. For this reason, for the Score-Driven models a Multivariate Student-t distribution with degrees of freedom $\nu > 2$ is chosen, so that the models may account for heavy tails in the data.

Furthermore, specifically for the Score-Driven models, the covariance matrix is adjusted as:

$$
\boldsymbol{H}_t = \sigma^2 \boldsymbol{I}_{N_t} + \boldsymbol{M}_t \boldsymbol{C} \boldsymbol{M}_t^\top \tag{42}
$$

where $\boldsymbol{C} \in \mathbb{R}^{p\times p}$ is an additional diagonal parameter matrix that captures uncertainty between prediction errors across the surface. This adjustment was proposed by Zou et al. (2025), who show that non-adjusted Score-Driven models performed substantially worse than their State-Space counterparts in density forecasts. This is motivated by the Kalman Filter 1-step ahead variance forecast given by $\boldsymbol{F}_t$ as defined in Equation (4):

$$
\boldsymbol{F}_t = \sigma^2 \boldsymbol{I}_{N_t} + \boldsymbol{M}_t \boldsymbol{P}_t \boldsymbol{M}_t^\top \tag{43}
$$

Therefore in a State-Space model the covariance forecast is non-diagonal even when the innovation covariance is diagonal. The formulation in Equation (42) thus mimics this behavior and was reported by the same authors to make the State-Space and Score-Driven formulations on par.

It was shown in Zou et al. (2025), that for $\nu > 2$, $\boldsymbol{\xi}_t$ is:

$$
\boldsymbol{\xi}_t = \boldsymbol{A}\boldsymbol{s}_t = \frac{1 + (N_t + 2)/\nu}{1 + \frac{\boldsymbol{\varepsilon}_t^\top \boldsymbol{H}_t^{-1} \boldsymbol{\varepsilon}_t}{\nu - 2}} \boldsymbol{A} \left(\boldsymbol{M}_t^\top \boldsymbol{H}_t^{-1} \boldsymbol{M}_t\right)^{-1} \boldsymbol{M}_t^\top \boldsymbol{H}_t^{-1} \boldsymbol{\varepsilon}_t \tag{44}
$$

where $\boldsymbol{A} \in \mathbb{R}^{p\times p}$ is a parameter matrix and $\boldsymbol{s}_t$ is the scaled score of the model. For $\nu \to \infty$, the score collapses to the score of a Gaussian Score-Driven model, in which the time-varying regression parameter $\boldsymbol{\beta}_t$ is adjusted using a GLS improvement step. For $2 < \nu < \infty$ the score down-weights the GLS step for large outliers through the $\boldsymbol{\varepsilon}_t^\top \boldsymbol{H}_t^{-1} \boldsymbol{\varepsilon}_t$ term in the denominator. Then in the Score-Driven model the parameter set is $\boldsymbol{\theta} = (\nu, \sigma^2, \mathrm{diag}(\boldsymbol{A})^\top, \mathrm{diag}(\boldsymbol{B})^\top, \mathrm{diag}(\boldsymbol{C})^\top, \bar{\boldsymbol{\beta}}^\top)^\top$ which can be estimated by directly maximizing the log-likelihood function:

$$\begin{aligned}\mathcal{L}(\boldsymbol{\theta}) = -\frac{1}{2}\sum_{t=1}^{T} \Bigg[& \log|(\nu-2)\pi \boldsymbol{H}_t| + (\nu + N_t)\log\left(1 + \frac{\boldsymbol{\varepsilon}_t^\top \boldsymbol{H}_t^{-1}\boldsymbol{\varepsilon}_t}{\nu - 2}\right) \\ & +2\left(\log\Gamma\left(\frac{\nu}{2}\right) - \log\Gamma\left(\frac{\nu+N_t}{2}\right)\right)\Bigg]\end{aligned} \tag{45}$$

Both of these specifications are called 'short memory' because the state $\boldsymbol{\beta}_t$ has exponential decay in both specifications.

3.2 Fractionally Integrated model

In the fractionally integrated models, the factor loadings $\boldsymbol{\beta}_t$ satisfy the following equation:

$$(1-L)^{\boldsymbol{d}}(\boldsymbol{\beta}_{t+1} - (\boldsymbol{I}_p - \boldsymbol{B})\bar{\boldsymbol{\beta}}) = \boldsymbol{B}(1-L)^{\boldsymbol{d}}\boldsymbol{\beta}_t + \boldsymbol{\xi}_t \tag{46}$$

where $\boldsymbol{d} \in \mathbb{R}^p$ is a vector of parameters with $d_i \in (-0.5, 0.5)$, $i = 1, \dots, p$ controlling the Long Range Dependence of the dimensions of $\boldsymbol{\beta}_t$, and L is the lag operator.

Specifically for IV surface data, preliminary empirical estimates of the autoregressive matrix $\boldsymbol{B}$ show that it is not significantly different from a $p \times p$ 0-matrix. This result may be interpreted as the fractional kernel being able to capture all the temporal dependence in the data.

Therefore, the model in Equation (46) can be simplified without loss of performance to:

$$(1-L)^{\boldsymbol{d}}(\boldsymbol{\beta}_{t+1} - \bar{\boldsymbol{\beta}}) = \boldsymbol{\xi}_t \tag{47}$$

As illustrated in Section 2.1, setting up a State-Space representation of this model would require running a state filter of the form of Equation (9), which increases in dimension at each t, which would require extensive computer memory complexity, even for small p. For this reason, only the adjusted Score-Driven specification is considered.

Then the dynamics of $\boldsymbol{\beta}_t$ can be rewritten as:

$$\begin{aligned}\boldsymbol{\beta}_{t+1} &= \bar{\boldsymbol{\beta}} + \sum_{\tau=0}^{\infty} \boldsymbol{W}_{FI}(\tau, \boldsymbol{d})\boldsymbol{A}\boldsymbol{s}_{t-\tau} \\ &\approx \bar{\boldsymbol{\beta}} + \sum_{\tau=0}^{t-1} \boldsymbol{W}_{FI}(\tau, \boldsymbol{d})\boldsymbol{A}\boldsymbol{s}_{t-\tau}\end{aligned} \tag{48}$$

where $\boldsymbol{\xi}_t = \boldsymbol{A}\boldsymbol{s}_t$ is defined as in Equation (44), the second line of Equation (48) is a finite-sample truncation, and the matrix-kernels $\boldsymbol{W}_{FI}(\tau, \boldsymbol{d})$ follow from the binomial expansion of $(1-L)^{-\boldsymbol{d}}$:

$$[\boldsymbol{W}_{FI}(\tau,\boldsymbol{d})]_{i,j} = \begin{cases} \frac{\Gamma(\tau+\boldsymbol{d}_i)}{\Gamma(\boldsymbol{d}_i)\Gamma(\tau+1)} & i=j \\ 0 & i\neq j \end{cases} \tag{49}$$

Note that unlike the short memory models, for a sample of size T the dynamics include increasingly many matrices $\boldsymbol{W}_{FI}(\tau,\boldsymbol{d})$, $\tau=1,\ldots,T-1$ governed by the fractional integration vector $\boldsymbol{d}$. Then the parameter vector $\boldsymbol{\theta}=(\sigma^2,\nu,\mathrm{diag}(\boldsymbol{A})^\top,\mathrm{diag}(\boldsymbol{C})^\top,\bar{\boldsymbol{\beta}}^\top,\boldsymbol{d}^\top)^\top$ can be estimated by maximizing the likelihood function in Equation (45).

3.3 Multifractal Score-Driven Model

This section develops a variant of the model which creates Long Range Dependence in a different way.

$\log \mathbf{IV}_t$ follows Equation (39). The factor loadings vector is decomposed as $\boldsymbol{\beta}_t = \left[\sigma_t,\, \tilde{\boldsymbol{\beta}}_t^\top\right]^\top$, where σ_t is the intercept loading. $\tilde{\boldsymbol{\beta}}_t$ follows an update as in Equation (40). The intercept loading is modeled as a Markov-Switching Multifractal (MSM) process of Calvet and Fisher (2004):

$$\sigma_t = \sigma_0 \prod_{k=1}^{K} G_{k,t}, \tag{50}$$

where $\sigma_0 \in \mathbb{R}^+$ is a static parameter and $G_{k,t}$, $k=1,\ldots,K$, are latent volatility components. Each $G_{k,t}$ is drawn independently from a fixed marginal distribution $\mathcal{G}$ with probability w_k, and otherwise remains at its current value:

$$G_{k,t} = \begin{cases} \text{draw from } \mathcal{G} & \text{with probability } w_k, \\ G_{k,t-1} & \text{with probability } 1-w_k, \end{cases} \tag{51}$$

with switching events independent across k and t. The marginal $\mathcal{G}$ is binomial-distributed, taking values $0<g_0<2$ or $2-g_0$ with equal probability, so that each $G_{k,t}$ has unit expectation and σ_0 is identified. The switching probabilities are specified as

$$w_k = 1-(1-w_K)^{\phi^{k-K}}, \quad \phi>1, \quad w_K\in(0,1), \tag{52}$$

so that $w_1<w_2<\cdots<w_K$: lower-indexed components switch less frequently and thus capture lower-frequency variation in the intercept, while higher-indexed components contribute short-lived fluctuations. Equation (52) implies that the logarithms of staying probabilities $\ln(1-w_k)$ are geometrically spaced at rate ϕ, which is the discrete-time analog of a Poisson arrival process with geometrically spaced intensities.

Then the intercept term is Multifractal in the sense that:

$$\mathbb{E}(|\sigma_{t+\tau}-\sigma_t|^q) \sim c(q)\tau^{\eta(q)+1} \tag{53}$$

where $\eta(q)$ and $c(q)$ are scaling functions and $\eta(q)$ is concave. The property in Equation 53 highlights one of the main differences between the fractionally integrated processes and the multifractal ones: while the moments of the FI processes have a power law scaling $\eta(q)$ that changes linearly in the moments q, in the multifractal models, this power law is nonlinear in q and strictly concave, leading to possibly more complex dynamics.

The vector $\mathbf{G}_t = (G_{1,t}, \ldots, G_{K,t})$ can be in $h = 2^K$ possible states $\{\boldsymbol{g}^{(1)}, \ldots, \boldsymbol{g}^{(h)}\} \in \mathbb{R}^{K+}$. Let $u(\boldsymbol{g}^{(j)}) = \prod_{k=1}^{K} g_k^{(j)}$, so that the intercept in state $\boldsymbol{g}^{(j)}$ equals $\sigma_0\, u(\boldsymbol{g}^{(j)})$. The transition matrix $\boldsymbol{\pi} \in \mathbb{R}^{h\times h}$ has entries

$$\boldsymbol{\pi}_{ij} = \prod_{k=1}^{K} \Big[(1-w_k)\, \mathbf{1}_{\{g_k^{(i)} = g_k^{(j)}\}} + w_k\, \mathbb{P}(\mathcal{G} = g_k^{(j)}) \Big] . \tag{54}$$

Denote by $\mathbf{\Pi}_{t|t-1} = (\Pi_{t|t-1}^{(1)}, \ldots, \Pi_{t|t-1}^{(h)})$ the vector of one-step-ahead predicted state probabilities, with $\mathbf{\Pi}_{t|t-1} = \mathbf{\Pi}_{t-1}\boldsymbol{\pi}$. The conditional density of $\log \mathbf{IV}_t$ given state $\boldsymbol{g}^{(j)}$ and $\tilde{\boldsymbol{\beta}}_t$ is:

$$p_{\nu,j} \equiv p_\nu\Big(\log \boldsymbol{IV}_t \mid \boldsymbol{G}_t = \boldsymbol{g}^{(j)},\, \tilde{\boldsymbol{\beta}}_t\Big) \tag{55}$$

In a State-Space specification, the Kalman filter does not provide efficient estimates and Monte Carlo methods would be required; therefore, only the adjusted Score-Driven update is considered. The marginal density integrates over all states:

$$p_\nu\Big(\log \boldsymbol{IV}_t \mid \tilde{\boldsymbol{\beta}}_t,\, \mathcal{F}_{t-1}\Big) = \sum_{j=1}^{h} \Pi_{t|t-1}^{(j)}\, p_{\nu,j} \tag{56}$$

The filtered state probabilities are updated via Bayes' rule as

$$\Pi_t^{(j)} = \frac{\Pi_{t|t-1}^{(j)}\, p_{\nu,j}}{\displaystyle\sum_{i=1}^{h} \Pi_{t|t-1}^{(i)}\, p_{\nu,i}}, \tag{57}$$

initialized at the ergodic distribution $\Pi_0^{(j)} = \prod_{k=1}^{K} \mathbb{P}(\mathcal{G} = g_k^{(j)})$, which follows from the independence of the components at time zero.

Appendix A shows that the score of the marginal log-density of $\log \mathbf{IV}_t$ with respect to $\tilde{\boldsymbol{\beta}}_t$ is:

$$\boldsymbol{s}_t = \big(\boldsymbol{M}_{t,2:p}^\top \boldsymbol{H}_t^{-1} \boldsymbol{M}_{t,2:p}\big)^{-1} \boldsymbol{M}_{t,2:p}^\top \boldsymbol{H}_t^{-1} \sum_{j=1}^{h} \Pi_t^{(j)} \frac{1 + (N_t+2)/\nu}{1 + \boldsymbol{\varepsilon}_{j,t}^\top \boldsymbol{H}_t^{-1} \boldsymbol{\varepsilon}_{j,t}/(\nu-2)} \boldsymbol{\varepsilon}_{j,t} \tag{58}$$

where $\boldsymbol{M}_{t,2:p}$ denotes the columns of $\boldsymbol{M}_t$ corresponding to $\tilde{\boldsymbol{\beta}}_t$. The filtered probabilities $\Pi_t^{(j)}$ in (58) are available from the forward filter step (57). The log-likelihood function is then:

$$\mathcal{L}(\boldsymbol{\theta}) = \sum_{t=1}^{T} \log \sum_{j=1}^{h} \Pi_{t|t-1}^{(j)}\, p_{\nu,j} \tag{59}$$

which can be used to estimate the parameter vector

$\boldsymbol{\theta} = \left(\nu,\, g_0,\, \sigma^2,\, \sigma_0,\, \phi,\, w_K,\, \mathrm{vec}(\boldsymbol{B})^\top, \mathrm{vec}(\boldsymbol{A})^\top, \mathrm{vec}(\boldsymbol{C})^\top, \bar{\boldsymbol{\beta}}^\top\right)$. The four MSM parameters $(g_0, \sigma_0, \phi, w_K)$ replace the fractional integration parameter $\boldsymbol{d}$. The number of frequency components K controls the richness of the volatility state space and is treated as a model selection problem, as is standard practice in multifractal estimation (Calvet and Fisher, 2004).

Note that unlike the fractionally integrated process, only the intercept of the factor model has a Long Range Dependence adjustment. This choice is motivated by the computational memory cost of the forward filter, which grows as 2^K in the naive procedure, due to the transition matrix size. Therefore, including a multifractal component per factor p would further increase storage requirements.

In fact, Calvet et al. (2006) found that using particle filters to approximate the joint distribution is more feasible than the analytical forward filter due to this storage requirement. A different approach is used here. These memory requirements can be reduced by leveraging the independence of the cascade components $G_{k,t}$, which allows to construct the filter in tensor form, reducing its complexity to $O(4K)$.

Since the K components switch independently, $\boldsymbol{\pi}$ admits an exact decomposition into K marginal transition matrices. For each $k = 1, \dots, K$, define

$$\boldsymbol{\pi}_k = (1 - w_k)\boldsymbol{I}_2 + w_k \mathbf{1}_2 \boldsymbol{p}_G^\top \in \mathbb{R}^{2\times 2}, \tag{60}$$

where $\boldsymbol{p}_G = [1/2,\, 1/2]^\top$ is the mass function of $\mathcal{G}$ and $\mathbf{1}_2 = [1, 1]^\top$. The K matrices are stored in a transition tensor $\tilde{\boldsymbol{\pi}} \in \mathbb{R}^{K\times 2\times 2}$ with $[\tilde{\boldsymbol{\pi}}]_k = \boldsymbol{\pi}_k$, requiring $O(4K)$ storage in place of $O(4^K)$. Intuitively, this reduction holds because, since the transitions are independent, explicitly storing the joint transition matrix does not carry any additional information.

The predicted probability vector $\boldsymbol{\Pi}_{t|t-1} \in \mathbb{R}^h$ is reshaped as a K-dimensional tensor $\tilde{\boldsymbol{\Pi}}_{t|t-1} \in \mathbb{R}^{2\otimes K}$, where dimension k indexes the two possible values of component k and $\otimes$ denotes the K-fold tensor product space. The joint transition is then replaced by K sequential mode-k products:

$$\boldsymbol{\Pi}_{t-1}\boldsymbol{\pi} = \mathrm{vec}\left[\tilde{\boldsymbol{\Pi}}_{t-1} \times_{k=1}^{K} \boldsymbol{\pi}_k\right], \tag{61}$$

where $\times_k$ denotes the mode-k tensor product. Each mode-k contraction costs $O(2 \cdot 2^K)$ operations, for a total of $O(K \cdot 2^{K+1})$ across all K modes, replacing the $O(4^K)$ matrix-vector product with no approximation error.

The observation weights $\{p_{\nu,j}\}_{j=1}^h$ defined in Equation (55) are evaluated at each t and arranged as a tensor $\tilde{\boldsymbol{p}}_{\nu,t} \in \mathbb{R}^{2\otimes K}$. Since $u(\boldsymbol{g}^{(j)}) = \prod_{k=1}^K g_k^{(j)}$ takes only $K+1$ distinct values $g_0^{K-l}(2-g_0)^l$, $l = 0, \dots, K$, the density $p_{\nu,j}$ needs to be evaluated at $K+1$ distinct intercepts $\sigma_0 g_0^{K-l}(2-g_0)^l$ rather than 2^K, with the remaining entries of $\tilde{\boldsymbol{g}}_t$ assigned by lookup. The Bayes update of equation (57), the prediction step,

Table 1: Memory and operation complexity of the matrix and tensor forward filters

	Matrix filter	Tensor filter
Transition storage	$O(4^K)$	$O(4K)$
Transition step	$O(4^K)$	$O(K \cdot 2^K)$
Density evaluations	$O(2^K)$	$O(K)$

Note: Reported computer memory storage requirements assume a MSM model with K components and binomial $\mathcal{G}$.

.

and the log-likelihood are then written as

$$\tilde{\mathbf{\Pi}}_t = \frac{\tilde{\mathbf{\Pi}}_{t|t-1} \odot \tilde{\boldsymbol{p}}_{\nu,t}}{\langle \tilde{\mathbf{\Pi}}_{t|t-1}, \tilde{\boldsymbol{p}}_{\nu,t} \rangle}, \tag{62}$$

$$\tilde{\mathbf{\Pi}}_{t+1|t} = \tilde{\mathbf{\Pi}}_t \times_{k=1}^{K} \boldsymbol{\pi}_k, \tag{63}$$

$$\mathcal{L}_t = \log \langle \tilde{\mathbf{\Pi}}_{t|t-1}, \tilde{\boldsymbol{p}}_{\nu,t} \rangle, \tag{64}$$

where $\odot$ denotes element-wise multiplication and $\langle \cdot, \cdot \rangle$ sums all elements of the element-wise product.

The score s_t of equation (58) is computed analogously. Let $\tilde{\boldsymbol{S}}_t \in \mathbb{R}^{(p-1)(2 \otimes K)}$ be the per-state scaled score tensor whose j-th element, under the flattening of Equation (61), equals the per-state gradient:

$$\tilde{\boldsymbol{S}}_t^{(j)} = \left(\boldsymbol{M}_{t,2:p}^\top \boldsymbol{H}_t^{-1} \boldsymbol{M}_{t,2:p} \right)^{-1} \boldsymbol{M}_{t,2:p}^\top \boldsymbol{H}_t^{-1} \frac{1 + (N_t + 2)/\nu}{1 + \boldsymbol{\varepsilon}_{j,t}^\top \boldsymbol{H}_t^{-1} \boldsymbol{\varepsilon}_{j,t} / (\nu - 2)} \boldsymbol{\varepsilon}_{j,t} \tag{65}$$

Since $\boldsymbol{\varepsilon}_{j,t}$ depends on the state only through $h(\boldsymbol{g}^{(j)})$, $\tilde{\boldsymbol{S}}_t$ also takes at most $K+1$ distinct values. Then:

$$\boldsymbol{s}_t = \sum_{j=1}^{2^K} \Pi_t^{(j)} \tilde{\boldsymbol{S}}_t^{(j)} \tag{66}$$

Table 1 reports the resulting complexity. The tensor representation reduces the cost of storing and applying $\boldsymbol{\pi}$, from $O(4^K)$ to $O(4K)$ in storage and from $O(4^K)$ to $O(K \cdot 2^K)$ in operations.

3.4 Markovian Long Range Dependence models

This section applies the method discussed in Section 2.2 to approximate Long Range Dependence while only using a first-order Markovian process. The factor loadings are specified as:

$$\begin{aligned} \boldsymbol{\beta}_{t+1} &= \bar{\boldsymbol{\beta}} + \sum_{i=0}^{K} \boldsymbol{W}_i \boldsymbol{b}_{i,t} \\ \boldsymbol{b}_{i,t+1} &= \boldsymbol{\mathcal{P}}_i \boldsymbol{b}_{i,t} + \sqrt{\boldsymbol{D}(\boldsymbol{I}_p - \boldsymbol{\mathcal{P}}_i^2) \boldsymbol{W}_i^{-1}} \boldsymbol{\xi}_t \end{aligned} \tag{67}$$

Where $\boldsymbol{\mathcal{P}}_i, \boldsymbol{W}_i \in \mathbb{R}^{p\times p} \forall i \in 0, ..., K$ are diagonal matrices with $\boldsymbol{\mathcal{P}}_i$ such that all of its eigenvalues are inside the unit circle, $\boldsymbol{b}_{i,t} \in \mathbb{R}^p$ are multivariate frequency components and $\boldsymbol{D}$ is an optional scaling factor to make the covariance comparable to other power laws as explained in Section 2.2 (here $\boldsymbol{D} = \boldsymbol{I}_p$ is assumed).

Similarly to the model of Section 3.2, the weighted sum of frequency components is designed to incorporate a power law decay in each of the components' auto-covariance. However, unlike the fractionally integrated process, the weights are finite dimensional and do not increase with time. In particular, let $\boldsymbol{\eta} \in \mathbb{R}^{p+}$ be the vector of the magnitude of power-law decays of each factor loading. The model components are specified by:

$$
\begin{aligned}
[\boldsymbol{\mathcal{P}}_i]_{k,j} &= \begin{cases} \exp(-\boldsymbol{\eta}_k \phi^{-i}) & k = j \\ 0 & k \neq j \end{cases} \\
\boldsymbol{c}_K &= \boldsymbol{I}_p \\
\boldsymbol{c}_{K-k} &= \boldsymbol{I}_p - \sum_{i=0}^{k-1} \boldsymbol{c}_{K-i} \phi^{-\boldsymbol{\eta}(k-i)} \exp\left(\boldsymbol{\eta}(1 - \phi^{-(k-i)})\right) \\
\boldsymbol{W}_i &= \boldsymbol{c}_i \phi^{-i\boldsymbol{\eta}} e^{\boldsymbol{\eta}}
\end{aligned}
\tag{68}
$$

where $\phi > 1$ is a scalar parameter that determines the geometric spacing between the target lags as explained in Section 2.2, $\phi^{\boldsymbol{\eta}}$ denotes the element wise exponentiation. When $\boldsymbol{\xi}_t$ defines a State-Space model, each frequency component follows an independent AR(1) recursion:

$$
\boldsymbol{b}_{i,t+1} = \boldsymbol{\mathcal{P}}_i \boldsymbol{b}_{i,t} + \sqrt{(\boldsymbol{I}_p - \boldsymbol{\mathcal{P}}_i^2)\boldsymbol{W}_i^{-1}} \boldsymbol{\xi}_{i,t}, \qquad \boldsymbol{\xi}_{i,t} \sim \mathcal{N}(\boldsymbol{0}, \boldsymbol{Q}),\ \boldsymbol{\xi}_{i,t} \perp \boldsymbol{\xi}_{j,t}\ \forall i \neq j, \tag{69}
$$

where $\boldsymbol{Q} \in \mathbb{R}^{p\times p}$ is a common diagonal innovation covariance. Stacking the frequency components into the augmented state $\tilde{\boldsymbol{b}}_t = (\boldsymbol{b}_{0,t}^\top, \ldots, \boldsymbol{b}_{K,t}^\top)^\top \in \mathbb{R}^{(K+1)p}$, the model admits the linear Gaussian state space form

$$
\tilde{\boldsymbol{b}}_{t+1} = \boldsymbol{\mathcal{P}}\, \tilde{\boldsymbol{b}}_t + \boldsymbol{\Phi}\tilde{\boldsymbol{\xi}}_t, \qquad \tilde{\boldsymbol{\xi}}_t \sim \mathcal{N}(\boldsymbol{0},\, \boldsymbol{I}_{K+1} \otimes \boldsymbol{Q})\,, \tag{70}
$$

$$
\log \boldsymbol{IV}_t = \boldsymbol{M}_t \bar{\boldsymbol{\beta}} + \tilde{\boldsymbol{M}}_t \tilde{\boldsymbol{b}}_t + \boldsymbol{\varepsilon}_t, \tag{71}
$$

where $\boldsymbol{\mathcal{P}} = \mathrm{diag}(\boldsymbol{\mathcal{P}}_0, \ldots, \boldsymbol{\mathcal{P}}_K)$, $\tilde{\boldsymbol{M}}_t = \boldsymbol{M}_t(\boldsymbol{W}_0 \mid \cdots \mid \boldsymbol{W}_K)$

and $\boldsymbol{\Phi} = \mathrm{blockdiag}\left(\sqrt{(\boldsymbol{I}_p - \boldsymbol{\mathcal{P}}_0^2)\boldsymbol{W}_0^{-1}}, \ldots, \sqrt{(\boldsymbol{I}_p - \boldsymbol{\mathcal{P}}_K^2)\boldsymbol{W}_K^{-1}}\right)$ is a block-diagonal matrix. Since the innovations $\boldsymbol{\xi}_{k,t}$ are independent across frequency components, the implied autocovariance of $\boldsymbol{\beta}_t$ is:

$$
\gamma(\tau) = \boldsymbol{Q} \sum_{i=0}^{K} \boldsymbol{W}_i \boldsymbol{P}_i^{\tau}, \tag{72}
$$

which matches the approximation as in Equation (31). Note that differently from the fractionally integrated model, the augmented state $\tilde{\boldsymbol{b}}_t$ is fixed-dimensional, therefore for a number of components much smaller than the sample size $K \ll T$, the state equation has much lower computer storage requirements.

Moreover, differently from the multifractal specification, the system is linear-Gaussian, so the Kalman filter applies exactly and can be used to estimate the parameter vector $\boldsymbol{\theta} = (\phi, \boldsymbol{\eta}^\top, \sigma^2, \mathrm{vec}(\boldsymbol{Q})^\top, \bar{\boldsymbol{\beta}}^\top)^\top$. In the (adjusted) Score-Driven model, the independent, Gaussian innovations in $\boldsymbol{\beta}_t$ are replaced by the scaled score with respect to each affine component $\boldsymbol{b}_t$ and the distribution becomes a multivariate Student's t distribution with the adjustment in Equation (42). From the chain rule the scaled score is $\boldsymbol{W}_i \boldsymbol{s}_t$, where $\boldsymbol{s}_t$ is defined as in Equation (44). To make the power law approximation work for the autocovariance, the algorithm in Equation (38) is employed. It then follows that the affine terms update is:

$$
\begin{aligned}
\boldsymbol{b}_{i,t+1} &= \boldsymbol{\mathcal{P}}_i\, \boldsymbol{b}_{i,t} + \boldsymbol{\kappa}_i \sqrt{(\boldsymbol{I}_p - \boldsymbol{\mathcal{P}}_i^2)\, \boldsymbol{W}_i}\; \boldsymbol{A}\, \boldsymbol{s}_t \\
\boldsymbol{L}_{i,j} &= \sqrt{\boldsymbol{W}_j \boldsymbol{W}_i^{-1}}\; \sqrt{(\boldsymbol{I}_p - \boldsymbol{\mathcal{P}}_i^2)(\boldsymbol{I}_p - \boldsymbol{\mathcal{P}}_j^2)}\; (\boldsymbol{I}_p - \boldsymbol{\mathcal{P}}_i \boldsymbol{\mathcal{P}}_j)^{-1} \\
\boldsymbol{\mu}_0^{(0)} &= \boldsymbol{I}_p \\
\boldsymbol{z}_i^{(0)} &= \sum_{j<i}^{K} \boldsymbol{L}_{i,j}\, \boldsymbol{\mu}_j^{(0)} \\
\boldsymbol{\mu}_i^{(0)} &= \frac{-\boldsymbol{z}_i^{(0)} + \sqrt{(\boldsymbol{z}_i^{(0)})^2 + 4\boldsymbol{I}_p}}{2}
\end{aligned}
\tag{73}
$$

$$
\begin{aligned}
\boldsymbol{z}_i^{(J)} &= \sum_{j\neq i}^{K} \boldsymbol{L}_{i,j}\, \boldsymbol{\mu}_j^{(J-1)}, \qquad J = 1, 2, \ldots \\
\boldsymbol{\mu}_i^{(J)} &= \frac{-\boldsymbol{z}_i^{(J)} + \sqrt{(\boldsymbol{z}_i^{(J)})^2 + 4\boldsymbol{I}_p}}{2} \quad \text{until } \|\boldsymbol{\mu}^{(J)} - \boldsymbol{\mu}^{(J-1)}\| < \epsilon \\
\boldsymbol{\kappa}_i &= \boldsymbol{\mu}_i^{(J)}\, \boldsymbol{W}_i^{-1}
\end{aligned}
\tag{74}
$$

where $\boldsymbol{\kappa}_i, \boldsymbol{z}_i^{(J)}, \boldsymbol{L}_{i,j}, \boldsymbol{\mu}_i^{(J)} \in \mathbf{R}^{p\times p}$ are diagonal matrices, and all powers, roots and inverses are applied entry-wise on the diagonal elements. Moreover, to make sure the procedure for finding the factors $\boldsymbol{\kappa}$ converges, in the Score-Driven version $\phi > 7.5$ is added as a heuristic constraint, as suggested in Section 2.2. The parameter vector $\boldsymbol{\theta} = (\nu, \phi, \boldsymbol{\eta}^\top, \sigma^2, \mathrm{diag}(\boldsymbol{A})^\top, \mathrm{diag}(\boldsymbol{C})^\top, \bar{\boldsymbol{\beta}}^\top)^\top$ is found by Maximizing the Likelihood function in Equation (45). If the model is fit using a gradient-based solver, it is more convenient to compute the terms κ_i after a fixed number of iterations of the algorithm in Equation (74), rather than explicitly searching for convergence, so that the computations become more stable.

4 Simulation Study

In this section, the out-of-sample performance of the different models, short memory State-Space (SS) and Score-Driven (SD), Fractionally Integrated Score-Driven model (FI-SD), Markov Switching Multifractal (MSM) and Markovian Long Range Dependence State-Space (Mk-SS) and Score-Driven (Mk-SD), is investigated in a Monte Carlo study. A time series of vector observations $\boldsymbol{y}_1, \ldots, \boldsymbol{y}_T$ is generated from the FI-SD model of Section 3.2 and then this specification is fit to the above specifications. The performance is evaluated by comparing log-likelihood, mean squared error (MSE), and mean absolute error (MAE) criterion for the one-step-ahead forecast so that both point and density forecasts are evaluated. Then multiple step-ahead point forecasts are evaluated at different horizons.

The factor loadings $\boldsymbol{M}_t$, include an intercept, moneyness levels $\mathbf{m}_t = (0.9,\ 0.98,\ 1.05,\ 1.15,\ 1.3,\ 1.5)$, and time-to-maturity levels $\mathbf{l}{=}(10,\ 50,\ 100,\ 180)/255$, resulting in a matrix of dimension 24×3 (the factors are augmented in Section 5). The choice of parameters reflects the estimates for the Option data in the empirical section. The innovation is drawn from a multivariate Student's t distribution with $\nu = 3$ degrees of freedom, unit observation noise variance $\sigma^2 = 1$, fractional integration parameters with high positive persistence $\boldsymbol{d} = (0.40, 0.40, 0.40)^\top$, scaling matrix $\boldsymbol{A} = \text{diag}(0.64, 0.61, 0.64)$, covariance adjustment matrix $\boldsymbol{C} = \text{diag}(0.10, 1.08, 0.10)$ and unconditional factors $\bar{\boldsymbol{\beta}} = (0.36, 0.90, 0.03)^\top$.

A time series of 4000 observations is generated, the first half of the observations is used to fit the models by Maximum Likelihood, and the second half is used for evaluation. In this way any potential biases that could be due to overfitting are avoided.

Table 2 presents the results for the one-step ahead forecast along with the significance of the Diebold-Mariano test statistics (Diebold and Mariano (2002)) with the Fractionally-Integrated Score-Driven model as a benchmark to quantify the statistical significance of the results. This table highlights 3 findings. First, both the multifractal models and the Markovian approximations have similar densities to their short memory alternatives, and they are statistically different from the fractionally integrated model's density at the 1% confidence level. Second, the Markovian Score-Driven approximating models are the only models for which the null-hypothesis of equal predictive ability from the DGP for MSE is failed to be rejected at the 10% confidence level. Interestingly, the same does not happen for the MAE, meaning that the Markovian long memory models have statistically indistinguishable performances in point forecasts to the DGP when more weight is given to large errors, but are statistically different when the central tendency is given more importance. Third, the number of components K does not particularly influence the 1-step ahead performance of the models. This is expected in the Markovian State-Space and Score-Driven models, since as seen in Section 2.2, similar parameter estimates will yield the same approximation of Long Range Dependence within their approxima-

Table 2: One-step-ahead predictive performance on simulated data.

Model	K	$\sigma^2 = 1$		
		MSE	MAE ($\times 10^2$)	loglik ($\times 10^{-3}$)
Mk-SD	1	2.651	$104.8^{\ddagger}$	$-57.76^{\ddagger}$
	2	2.649	$104.7^{\ddagger}$	$-57.75^{\ddagger}$
	3	2.648	$104.7^{\ddagger}$	$-57.74^{\ddagger}$
	5	2.647	$104.7^{\ddagger}$	$-57.74^{\ddagger}$
Mk-SS	1	$2.855^{\ddagger}$	$110.3^{\ddagger}$	$-73.59^{\ddagger}$
	2	$2.848^{\ddagger}$	$110.1^{\ddagger}$	$-73.58^{\ddagger}$
	3	$2.844^{\ddagger}$	$110.0^{\ddagger}$	$-73.57^{\ddagger}$
	5	$2.842^{\ddagger}$	$110.0^{\ddagger}$	$-73.57^{\ddagger}$
	10	$2.848^{\ddagger}$	$110.1^{\ddagger}$	$-73.58^{\ddagger}$
MSM	1	$2.671^{\ddagger}$	$105.7^{\ddagger}$	$-57.87^{\ddagger}$
	2	$2.671^{\ddagger}$	$105.7^{\ddagger}$	$-57.87^{\ddagger}$
	3	$2.671^{\ddagger}$	$105.7^{\ddagger}$	$-57.87^{\ddagger}$
	5	$2.671^{\ddagger}$	$105.7^{\ddagger}$	$-57.87^{\ddagger}$
	10	$2.671^{\ddagger}$	$105.7^{\ddagger}$	$-57.87^{\ddagger}$
SD		$2.671^{\ddagger}$	$105.7^{\ddagger}$	$-57.87^{\ddagger}$
SS		$2.874^{\ddagger}$	$111.3^{\ddagger}$	$-73.63^{\ddagger}$
FI-SD		2.638	103.9	-57.63

Note: Superscripts denote significance of Diebold-Mariano test statistics with 1-lag, HAC-covariance against DGP: $^{\circ}$10%, †5%, ‡1%.

tion boundary. The same argument does not hold for the MSM models, in which convergence to the power law autocovariance should improve with an increasing number of Cascade components. Instead, both the density and the point forecasts of the MSM models are almost identical to the short memory Score-Driven model, signaling that combining Random Multiplicative Cascades to the Score-Driven dynamics does not improve performance.

To further investigate how a different number of components affects the predictive performance of the Markovian approximations, multiple-horizon point forecasts are evaluated for the models on the same simulated data from the fractionally integrated Score-Driven model. Table 3 reports the results. Consistently with the 1-step ahead results, the MSM models are very close to the short-memory Score-Driven model, regardless of the number of components K and the horizon h. The Markovian Approximation models have the closest performances to the DGP, and for the State-Space models the Diebold-Mariano test statistic fails to reject the null hypothesis of equal predictive performance for all considered confidence levels, up to the 66-step

Table 3: Multi-horizon point forecast MSE

Model	K	$\sigma^2 = 1$			
		$h = 5$	$h = 22$	$h = 66$	$h = 260$
Mk-SD	1	$3.001^{\ddagger}$	$3.077^{\ddagger}$	$3.123^{\ddagger}$	$2.956^{\ddagger}$
	2	$2.992^{\ddagger}$	$3.065^{\ddagger}$	$3.116^{\ddagger}$	$2.956^{\ddagger}$
	3	$2.990^{\ddagger}$	$3.058^{\ddagger}$	$3.108^{\ddagger}$	$2.955^{\ddagger}$
	5	$2.986^{\ddagger}$	$3.053^{\ddagger}$	$3.100^{\ddagger}$	$2.950^{\ddagger}$
Mk-SS	1	$2.978^{\ddagger}$	$3.036^{\dagger}$	$3.105^{\ddagger}$	$2.962^{\ddagger}$
	2	$2.969^{\dagger}$	3.014	3.070°	$2.959^{\ddagger}$
	3	2.957	3.013	3.052	$2.917^{\ddagger}$
	5	2.961°	3.012	3.052	$2.926^{\ddagger}$
	10	2.956	3.011	3.051	$2.921^{\ddagger}$
MSM	1	$3.079^{\ddagger}$	$3.091^{\ddagger}$	$3.123^{\ddagger}$	$2.958^{\ddagger}$
	2	$3.079^{\ddagger}$	$3.091^{\ddagger}$	$3.123^{\ddagger}$	$2.958^{\ddagger}$
	3	$3.079^{\ddagger}$	$3.091^{\ddagger}$	$3.123^{\ddagger}$	$2.958^{\ddagger}$
	5	$3.079^{\ddagger}$	$3.091^{\ddagger}$	$3.123^{\ddagger}$	$2.958^{\ddagger}$
	10	$3.079^{\ddagger}$	$3.091^{\ddagger}$	$3.123^{\ddagger}$	$2.958^{\ddagger}$
SD		$3.079^{\ddagger}$	$3.091^{\ddagger}$	$3.123^{\ddagger}$	$2.958^{\ddagger}$
SS		$3.035^{\ddagger}$	$3.089^{\ddagger}$	$3.122^{\ddagger}$	$2.961^{\ddagger}$
FI-SD		2.937	3.010	3.051	2.898

Note: Each column reports the MSE for the h-steps ahead point forecasts. Superscripts denote significance of Diebold-Mariano test statistics against DGP: $^{\circ}$10%, †5%, ‡1%. HAC-covariances' lags used are equal to h.

ahead forecasts, with performance deteriorating only at the 260-steps horizon.

5 Empirical Data and Forecast results

5.1 Descriptives

The dataset comprises European options on the S&P 500 index and encompasses all call and put options traded on the Chicago Board Options Exchange (CBOE). The dataset, retrieved from OptionsDX.com, spans the period from January 1, 2010, to January 1, 2024. It includes the daily closing price of the index, as well as the strike prices, expiration dates, option deltas (Δ), and implied volatilities of each option contract.

The same data cleaning procedure as in Zou et al. (2025), van der Wel et al. (2016), and Barone-Adesi et al. (2008) is applied. Only Out-of-The-Money (OTM) options with $|\Delta| < 0.5$ are considered due to their higher trading activity with

Table 4: Summary statistics by moneyness and maturity bucket.

		7–45 days		45–90 days		90–180 days		180–360 days	
		Mean	Std	Mean	Std	Mean	Std	Mean	Std
DOTM put	IV	0.33	0.13	0.33	0.12	0.35	0.11	0.36	0.11
	Moneyness	0.84	0.09	0.76	0.12	0.68	0.14	0.58	0.15
	TTM	22.88	11.03	64.68	12.37	126.91	26.28	266.24	52.23
OTM put	IV	0.20	0.08	0.22	0.07	0.24	0.06	0.24	0.06
	Moneyness	0.96	0.02	0.94	0.03	0.91	0.04	0.87	0.06
	TTM	25.07	10.45	65.87	12.66	131.52	26.77	271.99	52.89
ATM put	IV	0.18	0.08	0.18	0.07	0.20	0.06	0.20	0.05
	Moneyness	0.99	0.01	0.99	0.01	0.99	0.01	0.99	0.02
	TTM	24.91	10.35	65.92	12.73	132.33	26.97	272.01	53.23
ATM call	IV	0.17	0.08	0.17	0.07	0.18	0.05	0.18	0.04
	Moneyness	1.01	0.01	1.01	0.01	1.02	0.02	1.04	0.03
	TTM	24.70	10.42	65.89	12.71	132.16	26.86	273.16	53.03
OTM call	IV	0.15	0.08	0.15	0.06	0.16	0.05	0.16	0.04
	Moneyness	1.03	0.02	1.05	0.02	1.07	0.03	1.11	0.05
	TTM	24.32	10.65	65.86	12.64	129.30	26.35	272.03	52.54
DOTM call	IV	0.18	0.10	0.15	0.06	0.16	0.05	0.15	0.04
	Moneyness	1.11	0.10	1.16	0.12	1.23	0.15	1.33	0.18
	TTM	21.88	10.82	64.57	12.44	126.63	26.10	264.30	52.26

Note: This table presents summary statistics for the option data, including the mean and standard deviation (Std) over time for the implied volatility (IV), Time to maturity (TTM), moneyness (the strike price divided by the index) across four maturity groups and six moneyness groups. The maturity groups are 7–45, 45–90, 90–180, and 180–360 days. The six moneyness groups are defined as deep out-of-the-money put ($-0.125 < \Delta < 0$, DOTM put), out-of-the-money put ($-0.375 < \Delta < -0.125$, OTM put), at-the-money put ($-0.5 < \Delta < -0.375$, ATM put), and similarly for call options (with positive Δs). Each day, all contracts that fall within each maturity-moneyness group are identified, and the numbers represent averages over time and across contracts for each group.

respect to in-the-money quotes. Under put-call parity this is conceptually equivalent to studying only in-the-money options. Options with less than 7 or more than 360 days to expiration are excluded to avoid low liquidity level effects. Implied volatilities with more than 0.7 or less than 0.05 are also excluded to avoid potential data errors. The final dataset comprises a total of 12,193,386 observations, with an average of 3,493 observations per day.

Table 4 shows descriptive statistics of the IV surface. Following from Zou et al. (2025) and van der Wel et al. (2016), the sample is divided into 24 groups based on moneyness and time-to-maturity. Specifically, the maturity component is partitioned into four groups with breakpoints at 45, 90, and 180 days-to-maturity, while moneyness is split into six groups with breakpoints at Δ values of -0.375, -0.125, 0, 0.125, and 0.375. Options with Δ values ranging from -0.125 to 0 are classified as deep out-of-the-money puts (DOTM puts). Options with Δ from -0.375 to -0.125 are classified as out-of-the-money puts (OTM puts), and options with Δ values between -0.5 and -0.375 are classified as at-the-money puts (ATM puts). Calls are classified as deep OTM, OTM, and ATM, using the same cutoffs, but with positive Δ values. Implied volatilities decrease as moneyness increases for each of the four maturity groups, a phenomenon commonly known as the volatility skew or smirk. Second, the implied volatilities increase as the time-to-maturity increases, known as the volatility term structure.

5.2 Forecast

In this section, the main empirical results are presented. All analyses are performed out of sample. The first half of the observations are used for fitting the data, the second half is used for evaluation, giving $T = 1745$ for the 1-step ahead evaluation and $T - h$ for the h-steps ahead forecast evaluation. As in the simulation study, the one step ahead forecast includes both density and point estimates, and also includes the Akaike Information Criterion (AIC) and BIC (Bayesian Information Criterion) to account for the different number of parameters between the models. The multi-step-ahead forecasts focus on the Mean Squared Error (MSE). To assess the statistical significance of the results, the Model Confidence Set (MCS) of Hansen et al. (2011) is used, giving the set of models with the same predictive performance at different confidence levels for each of the point and density forecast statistics.

The models used are the same as in Sections 3 and 4, with the difference that the factors $\boldsymbol{M}_t$ are augmented by a fourth nonparametric term ω_x estimated by data and changing depending on which of the moneyness-maturity buckets the option quote is in. The DOTM-put, 7-45 days bucket is set to be the reference bucket $x = 1$ with $\omega_1 = 0$, and the other 23 bucket parameters $\omega_2, \cdots, \omega_{24}$ are estimated by maximum likelihood along with the rest of the parameters. Table 5 reports the one-step-ahead forecast results. Consistently with the Monte-Carlo study and Zou et al. (2025), the Student's t-distributed Score-Driven models consistently outperform the State-Space models in terms of density forecasts. Unlike the simulation study however, the short-memory Score-Driven model has the best density estimate, with the Markovian model with $K = 1$ as the second best. Although the multifractal and fractionally integrated models still outperform the State-Space specifications and are somewhat statistically relevant in some cases, their performance is slightly worse than the short-memory specifications. The AIC and BIC model confidence sets are identical to the log-likelihood results.

For the point forecast, the opposite happens: the State-Space specifications have

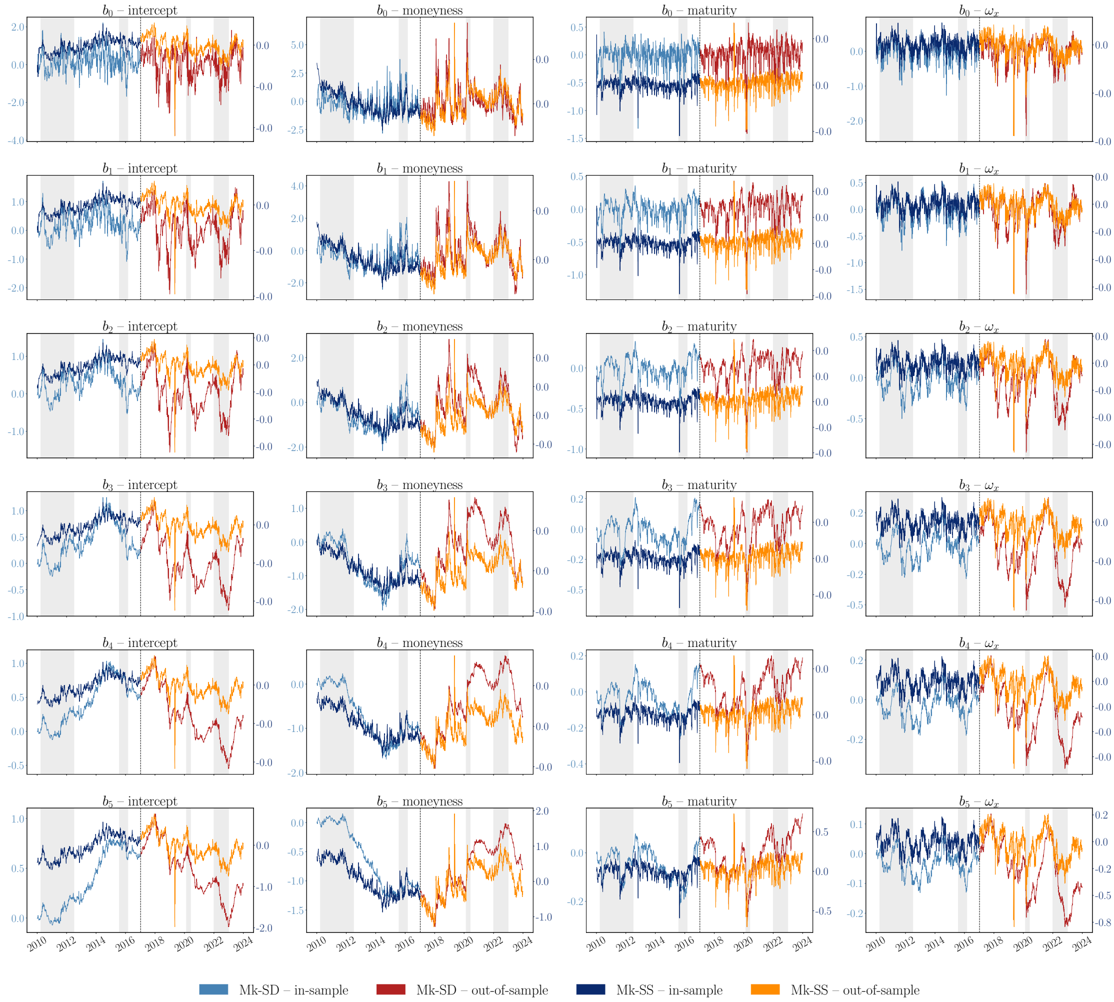
Table 5: One-step-ahead predictive performance

Model	K	MSE	MAE	Neg. Log-Lik.	AIC	BIC
Mk-SD	1	0.0500‡	0.1463	-856.0715‡	-1712.0963‡	-1711.7666‡
	2	0.0500‡	0.1464	-856.0635‡	-1712.0800‡	-1711.7502‡
	3	0.0500‡	0.1464	-856.0673‡	-1712.0879‡	-1711.7581‡
	5	0.0500‡	0.1464	-856.0681‡	-1712.0890‡	-1711.7594‡
	10	0.0500‡	0.1464	-856.0704‡	-1712.0936‡	-1711.7642‡
Mk-SS	1	0.0487‡	0.1439	-692.4225	-1384.8037	-1384.5143
	2	0.0487‡	0.1438‡	-692.4182	-1384.7952	-1384.5056
	3	0.0487‡	0.1438‡	-692.3788	-1384.7166	-1384.4270
	5	0.0487‡	0.1438	-692.4115	-1384.7817	-1384.4922
	10	0.0487‡	0.1438‡	-692.4320	-1384.8228	-1384.5333
MSM	1	0.0526	0.1524	-854.6280	-1709.2089	-1708.8793
	2	0.0512°	0.1489	-854.9426°	-1709.8383°	-1709.5085°
	3	0.0506†	0.1477	-854.1962	-1708.3455	-1708.0159
	5	0.0499‡	0.1457	-854.4839†	-1708.9208†	-1708.5912†
	10	0.0492‡	0.1455	-855.7809†	-1711.5150†	-1711.1853†
SD		0.0500‡	0.1458	-856.0738‡	-1712.1006‡	-1711.7710‡
SS		0.0487‡	0.1438‡	-692.6245	-1385.2076	-1384.9182
FI-SD		0.0534°	0.1535	-855.1821†	-1710.3174†	-1709.9877†

Note: Superscripts denote inclusion in MCS at different confidence levels: $^\circ$ 5%, †10%, ‡ 25%. HAC covariance and block-bootstrap use 10 lags. Elimination rule uses maximum t-statistics. Bootstrap uses 10000 simulations.

the lowest MSE and MAE across all models, with little difference between the short-memory and Markovian long memory specifications, which are the only models to remain in the MAE confidence set. The short-memory and Markovian long-memory Score-Driven models have almost identical performance, with both of them being in the MSE model confidence set at the 25% confidence level. The Multifractal models with a higher number of components also enter the model confidence set, and for $K = 5$ and $K = 10$ their error is slightly lower than the rest of the Score-Driven models. The fractionally integrated model is the worst performing model in terms of point forecast, and it remains in the model confidence set only at the 5% confidence level. Note that although the MSM model with $K = 1$ has a lower MSE than the FI model, the former is excluded because its MSE has very low variance, whereas the latter remains due to higher variance, which decreases its t-statistic in the MCS elimination rule. From these results, it becomes clear that for 1-step ahead forecast,

Figure 4: Markovian models' $\boldsymbol{b}_{i,t}$ over time



Note: Each panel plots the autoregressive components approximating long memory for the Markovian models with 5 components. Mk-SD's $\boldsymbol{b}_{i,t}$ are scaled on the left y-axes, while the State-Space models are scaled on the right y-axes. Dashed, vertical line denotes end of in-sample period. Background gray areas denote crises period, including - 2010-04 to 2012-07 Euro debt crisis - 2015-08 to 2016-02 — China/EM selloff - 2020-02 to 2020-06 — COVID crash - 2022-01 to 2022-12 — 2022 rate hike bear market.

the short-memory Score-Driven model is sufficient to capture the dynamics of the IV surface, because it is in both the density and point forecast model confidence set. The Markovian approximation models and the MSM have similar performances, while the fractionally integrated model is slightly worse, although not significantly so. This suggests that for short-term evaluations, the surface is indistinguishable from an exponentially decaying process, and using more complex models such as MSM, the fractional integrated model, or even the Markovian approximations may

Table 6: MSE across forecast horizons

Model	K	$h=5$	$h=22$	$h=66$	$h=260$
Mk-SD	1	$0.1012^{\ddagger}$	$0.1337^{\ddagger}$	$0.1597^{\ddagger}$	$0.1700^{\ddagger}$
	2	$0.1012^{\ddagger}$	$0.1335^{\ddagger}$	$0.1613^{\ddagger}$	0.1877
	3	$0.1013^{\ddagger}$	$0.1336^{\ddagger}$	$0.1612^{\ddagger}$	0.1869
	5	$0.1012^{\ddagger}$	$0.1334^{\ddagger}$	$0.1609^{\ddagger}$	0.1861°
	10	$0.1012^{\ddagger}$	$0.1334^{\ddagger}$	$0.1610^{\ddagger}$	0.1860°
Mk-SS	1	$0.1010^{\ddagger}$	$0.1358^{\ddagger}$	$0.1611^{\ddagger}$	$0.1671^{\ddagger}$
	2	$0.1012^{\ddagger}$	$0.1382^{\ddagger}$	$0.1732^{\dagger}$	0.2145
	3	$0.1012^{\ddagger}$	$0.1373^{\ddagger}$	$0.1647^{\ddagger}$	$0.1814^{\ddagger}$
	5	$0.1011^{\ddagger}$	$0.1363^{\ddagger}$	$0.1620^{\ddagger}$	$0.1711^{\ddagger}$
	10	$0.1010^{\ddagger}$	$0.1358^{\ddagger}$	$0.1608^{\ddagger}$	$0.1693^{\ddagger}$
MSM	1	$0.1034^{\dagger}$	$0.1352^{\ddagger}$	$0.1611^{\ddagger}$	$0.1705^{\ddagger}$
	2	$0.1027^{\dagger}$	$0.1353^{\ddagger}$	$0.1611^{\ddagger}$	$0.1677^{\ddagger}$
	3	$0.1025^{\dagger}$	$0.1338^{\ddagger}$	$0.1575^{\ddagger}$	$0.1658^{\ddagger}$
	5	$0.1018^{\ddagger}$	$0.1390^{\ddagger}$	0.1905	0.2974
	10	$0.1013^{\ddagger}$	$0.1355^{\ddagger}$	$0.1636^{\ddagger}$	$0.1710^{\ddagger}$
SD		$0.1015^{\ddagger}$	$0.1411^{\dagger}$	0.1950	0.4067
SS		$0.1012^{\ddagger}$	$0.1374^{\ddagger}$	$0.1664^{\ddagger}$	$0.1798^{\ddagger}$
FI-SD		$0.1122^{\dagger}$	$0.1416^{\ddagger}$	$0.1636^{\ddagger}$	$0.1751^{\ddagger}$

Note: Superscripts denote inclusion in MCS at different confidence levels: $^{\circ}$ 5%, † 10%, ‡ 25%. HAC covariance and block-bootstrap use 10 lags. Elimination rule uses maximum t-statistics. Bootstrap uses 10000 simulations.

be unnecessary.

To investigate this further, Figure 4 visualizes the filtered and 1-step ahead forecast of the $\boldsymbol{b}_{i,t}$ of Markovian approximating models with $K = 5$ for both the State-Space and Score-Driven specifications. For both of the models $\boldsymbol{b}_i$s have very different trajectories between the intercept, moneyness, maturity and ω_x dimension, while they are consistent within lower and higher frequency components on the same dimension, with higher frequency components being more stable. In the Score-Driven specification, the rows follow a clear hierarchy from noisy to smooth. $\boldsymbol{b}_0$ and $\boldsymbol{b}_1$ (top rows) are extremely erratic, with intercepts oscillating between 2 and -4 and moneyness coefficients varying between 6 and - 3. By $\boldsymbol{b}_3$–$\boldsymbol{b}_5$ the series become slow-moving cycles, and the scale shrinks substantially at each row. This is expected, as higher frequency components are required to approximate Long Range Dependence at higher lags, requiring higher persistence. The intercept and moneyness carry most of the variation. The ω_x column behaves like a secondary moneyness effect: same sign pattern, but smaller.

While a similar trajectory is observed for the State-Space specification, its first 4 components are very close to 0 throughout the whole period, with most of the variation concentrated to the fifth component. Along with the forecast results, this signals that the reason why the short-memory and Markovian long-memory specifications have such close performance is that on the Implied Volatility surface data the Markovian models are not trying to replicate Long Range Dependence, but rather to get as close as possible to the short-memory model.

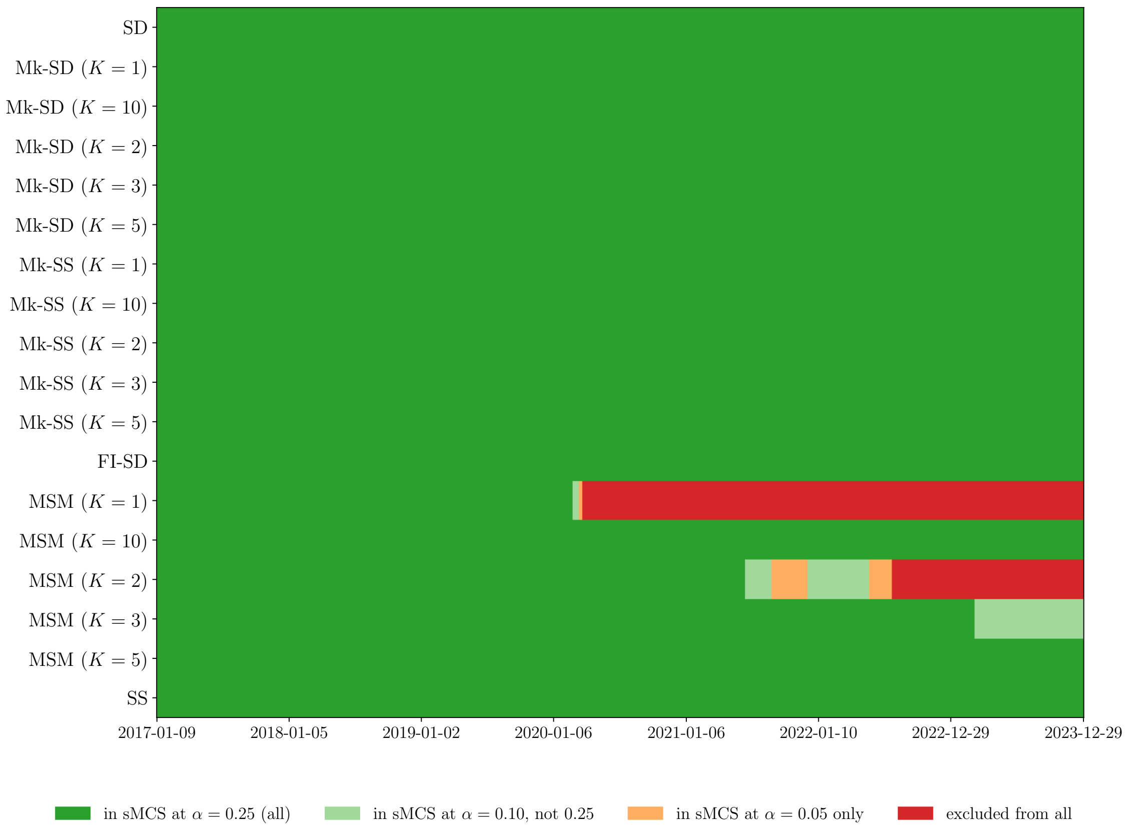
These results are also confirmed by the multiple horizon point forecasts. Table 6 presents the results. The Score-Driven and State-Space specifications are statistically equivalent for most of the models in the 5 and 22 step ahead forecasts, again suggesting that the short-memory specifications are sufficient to describe the Implied Volatility surface, since the fractionally integrated model and the MSM models with a higher number of components are not statistically better than the short-memory models. For the 66 and especially the 260 forecasting horizons some of the models are excluded from the model confidence set, with the short-memory models having statistically indistinguishable performances from the long memory models.

5.3 Robustness check with Sequential Model Confidence Set

To verify the robustness of the empirical findings, it would be desirable to see if the models are found to have the same predictive performance over time, not only on average. For example, Zou et al. (2025) also tests the out of sample performance of the models by splitting the COVID and the pre-COVID periods, to test how the models behave in a stress period. One possibility is to repeat the model confidence set procedure of Hansen et al. (2011) on the loss differentials separately on each of the sub-samples of interests. However, this would inflate the probability of wrongly excluding at least one model with superior predictive accuracy. Moreover, this error increases even more if the sub-periods are chosen after seeing the data, since the sample size of the losses is assumed to be independent of the data itself.

To overcome these issues, the Sequential Model Confidence Set procedure of Arnold et al. (2026) is used. In particular, the *weakly superior* set is constructed, in which starting from a set of models $\mathcal{M}_0$ and a loss function (e.g. MSE), for the out-of-sample time points $t = 1, \dots, T$, the procedure identifies a confidence *sequence* $\{\mathcal{M}_t^\star\}_{t=1}^T$ such that $\mathcal{M}_t^\star \subseteq \mathcal{M}_0$ contains the models whose expected loss differential at time t is smaller or equal to 0 for a specific confidence level. Unlike the traditional Model Confidence Set, a model can leave and re-enter the confidence sequence at different times (see Arnold et al. (2026) for technical and robustness details).
Figure 5 reports the inclusions of the models in the confidence sequence for the 1-step ahead MSE. All models are included in the confidence sequence for the first half of the out-of-sample period, and only the MSM models with less components are excluded from the second half of the sample, with the 1 component model being excluded from the beginning of 2020 and the 2 component model starting from

Figure 5: 1-step ahead MSE Model Confidence Sequence Inclusions



Note: Sequential model confidence sets (sMCS) under the weak hypothesis. At each time t, model i is included if there is no sufficient evidence that its cumulative average conditional loss exceeds any competitor's. Colors indicate membership at different Confidence Levels (α): dark green = in sMCS at all three levels ($\alpha \leq 0.25$); light green = $\alpha \leq 0.10$; orange = $\alpha \leq 0.05$ only; red = excluded from all. The procedure provides a time-uniform family-wise error guarantee simultaneously over all t; models may re-enter the set as cumulative evidence evolves. E-processes use a fixed per-pair betting parameter $\lambda_{ij} = (2c_{ij})^{-1}$ with $c_{ij} = 2\max_t |d_{ij,t}|$ and $d_{ij,t}$ the MSE differential between model i and j at time t.

the second half of 2022. Other confidence sequences for the multi-steps-ahead are available in the appendices, and are consistent with these results. This result aligns with Table 5, indicating that the volatility surface is modeled effectively by the short-memory models, and that including Long Range Dependence features is not necessary for forecasting accuracy.

6 Conclusion

Fractional integration and Random Multiplicative Cascades are the most common ways to generate processes with power law decay in the autocovariance function, but may introduce computational challenges with respect to short-memory models. To overcome these computational requirements, this thesis shows how to apply an analytical approximation procedure of an arbitrary power law to a first-order Markovian process, so that its autocovariance mimics Long Range Dependence parsimoniously. Using this procedure, the approximating Markovian models are shown to be able to recover the out of sample power law behavior of fractionally integrated processes without facing the computational requirements required by these models.

The base approximation can be used directly in State-Space models but does not work in their Score-Driven counterparts due to the common mechanism driving the update over time. Therefore, as a second contribution, an iterative procedure is proposed to make the approximation work in the Score-Driven setting, so that non-Gaussian innovations can be introduced without requiring simulation-based inference. The introduction of this second adjustment makes the approximation general enough to be applicable to both Score-Driven and State-Space models of different forms.

The new Markovian approximation is applied to dynamic factor models. A Monte Carlo study confirms that the approximating, Markovian models outperform their short-memory counterparts when the DGP has long memory at both short and moderate forecasting horizons. Instead, when the models are applied to the implied Volatility surface for the S&P 500 index options data, forecasting accuracy is found to be statistically indistinguishable between the short-memory, long-memory and Markovian approximating models, suggesting that the IV curve is not a long memory process.

Therefore, the natural conclusion is that the proposed adjustment provides a valid alternative to fractionally integrated models, and can be applied to high dimensional cases if the number of Long Range dependent objects is limited, as illustrated in the empirical section on implied Volatility surfaces, in which case one has more than hundreds of time series observations in each period. However, for modeling the S&P 500 Implied Volatility curve specifically, the empirical results suggest that the short-memory models suffice to explain its dynamics.

Further research could address whether this approximation works in other modeling contexts beyond the factor model context, or if the Implied Volatility surface of different assets can instead benefit from introducing Long Range Dependence components. This is left for future research.

References

Abi Jaber, E. (2019). Lifting the heston model. *Quantitative finance*, 19(12):1995–2013.

Arnold, S., Gavrilopoulos, G., Schulz, B., and Ziegel, J. (2026). Sequential model confidence sets. *Journal of the Royal Statistical Society Series B: Statistical Methodology*, page qkag066.

Baillie, R. T., Bollerslev, T., and Mikkelsen, H. O. (1996). Fractionally integrated generalized autoregressive conditional heteroskedasticity. *Journal of Econometrics*, 74:3–30.

Barone-Adesi, G., Engle, R. F., and Mancini, L. (2008). A garch option pricing model with filtered historical simulation. *The Review of Financial Studies*, 21(3):1223–1258.

Black, F. and Scholes, M. S. (1973). The pricing of options and corporate liabilities. *Journal of Political Economy*, 81:637 – 654.

Bochud, T. and Challet, D. (2007). Optimal approximations of power laws with exponentials: application to volatility models with long memory. *Quantitative Finance*, 7:585–589.

Bouchaud, J.-P., Potters, M., and Meyer, M. (2000). Apparent multifractality in financial time series. *The European Physical Journal B*, 13:595–599.

Buccheri, G. and Corsi, F. (2019). Hark the shark: Realized volatility modeling with measurement errors and nonlinear dependencies*. *Journal of Financial Econometrics*.

Calvet, L. E. and Fisher, A. J. (2002). Multifractality in asset returns: Theory and evidence. *The Review of Economics and Statistics*, 84:381–406.

Calvet, L. E. and Fisher, A. J. (2004). How to forecast long-run volatility: Regime switching and the estimation of multifractal processes. *Journal of Financial Econometrics*, 2:49–83.

Calvet, L. E., Fisher, A. J., and Thompson, S. B. (2006). Volatility comovement: a multifrequency approach. *Journal of Econometrics*, 131(1):179–215.

Challet, D. and Ragel, V. (2024). Multi-timescale recurrent neural networks beat rough volatility for intraday volatility prediction. *Risks*, 12:84.

Chan, N. H. and Palma, W. (1998). State space modeling of long-memory processes. *The Annals of Statistics*, 26(2):719–740.

Chronopoulou, A. and Spiliopoulos, K. (2018). Sequential monte carlo for fractional stochastic volatility models. *Quantitative Finance*, 18(3):507–517.

Conrad, C. and Karanasos, M. (2006). The impulse response function of the long memory garch process. *Economics Letters*, 90(1):34–41.

Corsi, F. (2009). A simple approximate long-memory model of realized volatility. *Journal of Financial Econometrics*, 7:174–196.

Cox, D. R. (1981). Statistical analysis of time series: Some recent developments. *Scandinavian Journal of Statistics*, 8(2):93–115.

Creal, D., Koopman, S. J., and Lucas, A. (2013). Generalized autoregressive score models with applications. *Journal of Applied Econometrics*, 28(5):777–795.

Diebold, F. X. and Mariano, R. S. (2002). Comparing predictive accuracy. *Journal of Business & Economic Statistics*, 20(1):134–144.

Durbin, J. and Koopman, S. J. (2012). *Time Series Analysis by State Space Methods*. OUP Oxford.

Granger, C. W. J. and Joyeux, R. (1980). An introduction to long-memory time series models and fractional differencing. *Journal of Time Series Analysis*, 1:15–29.

Hansen, P. R., Lunde, A., and Nason, J. M. (2011). The model confidence set. *Econometrica*, 79(2):453–497.

Hosking, J. R. M. (1981). Fractional differencing. *Biometrika*, 68:165–176.

Hualde, J. and Robinson, P. (2007). Root-n-consistent estimation of weak fractional cointegration. *Journal of Econometrics*, 140(2):450–484.

Johansen, S. and Nielsen, M. (2012). Likelihood inference for a fractionally cointegrated vector autoregressive model. *Econometrica*, 80(6):2667–2732.

LeBaron, B. (2001). Stochastic volatility as a simple generator of apparent financial power laws and long memory. *Quantitative Finance*, 1(6):621–631.

Shu, H. and Nan, B. (2019). Estimation of large covariance and precision matrices from temporally dependent observations. *The Annals of Statistics*, 47:1321–1350.

van der Wel, M., R. Ozturk, S., and van Dijk, D. (2016). Dynamic factor models. *Advances in Econometrics*, 35.

Zhang, Q., Park, C., and Chung, J. (2021). Minimax estimation of covariance and precision matrices for high-dimensional time series with long-memory. *Statistics Probability Letters*, 177:109177.

Zou, X., Lin, Y., and Lucas, A. (2025). Improving score-driven density forecasts with an application to implied volatility surface dynamics. *Tinbergen Institute Discussion Paper*.

A Derivation of Tensor Score

Since $\boldsymbol{G}_t$ is latent with predicted probability vector $\boldsymbol{\Pi}_{t|t-1}$ the conditional density of $\log \boldsymbol{IV}_t$ given $\tilde{\boldsymbol{\beta}}_t$ and $\mathcal{F}_{t-1}$ is the mixture

$$p_\nu\Big(\log \boldsymbol{IV}_t \mid \tilde{\boldsymbol{\beta}}_t, \mathcal{F}_{t-1}\Big) = \sum_{j=1}^{h} \Pi^{(j)}_{t|t-1}\, p_{\nu,j}, \tag{75}$$

where

$$p_{\nu,j} = \frac{\Gamma\left(\frac{\nu+N_t}{2}\right)}{\Gamma\left(\frac{\nu}{2}\right)\left(\pi(\nu-2)\right)^{N_t/2}|\boldsymbol{H}_t|^{1/2}}\left(1+\frac{\boldsymbol{\varepsilon}_{j,t}^\top \boldsymbol{H}_t^{-1}\boldsymbol{\varepsilon}_{j,t}}{\nu-2}\right)^{-\frac{\nu+N_t}{2}} \tag{76}$$

and

$$\boldsymbol{\varepsilon}_{j,t} = \log \boldsymbol{IV}_t - \boldsymbol{M}_t \begin{pmatrix} \sigma_0\, u(\boldsymbol{g}^{(j)}) \\ \tilde{\boldsymbol{\beta}}_t \end{pmatrix} \tag{77}$$

where definitions are as in Section 3.3. The raw score of the log-mixture with respect to $\tilde{\boldsymbol{\beta}}_t$ is:

$$\begin{aligned}
\boldsymbol{\nabla}_t &\equiv \frac{\partial}{\partial \tilde{\boldsymbol{\beta}}_t} \log \sum_j \Pi^{(j)}_{t|t-1}\, p_{\nu,j} \\
&= \frac{\sum_j \Pi^{(j)}_{t|t-1}\, \partial p_{\nu,j}/\partial \tilde{\boldsymbol{\beta}}_t}{\sum_j \Pi^{(j)}_{t|t-1}\, p_{\nu,j}} \\
&= \sum_j \frac{\Pi^{(j)}_{t|t-1}\, p_{\nu,j}}{\sum_i \Pi^{(i)}_{t|t-1}\, p_{\nu,i}} \cdot \frac{\partial \log p_{\nu,j}}{\partial \tilde{\boldsymbol{\beta}}_t} \\
&= \sum_j \Pi^{(j)}_t\, \boldsymbol{\nabla}_j,
\end{aligned} \tag{78}$$

where the Bayes update $\Pi^{(j)}_t = \Pi^{(j)}_{t|t-1} p_{\nu,j} / \sum_i \Pi^{(i)}_{t|t-1} p_{\nu,i}$ is used in the last equality. The per-state score is

$$\begin{aligned}
\boldsymbol{\nabla}_j \equiv \frac{\partial \log p_{\nu,j}}{\partial \tilde{\boldsymbol{\beta}}_t} &= -\frac{\nu+N_t}{2} \cdot \frac{2/(\nu-2)}{1+\boldsymbol{\varepsilon}_{j,t}^\top \boldsymbol{H}_t^{-1}\boldsymbol{\varepsilon}_{j,t}/(\nu-2)} \cdot \left(-\boldsymbol{M}_{t,2:p}^\top \boldsymbol{H}_t^{-1}\boldsymbol{\varepsilon}_{j,t}\right) \\
&= \frac{\nu+N_t}{\nu-2+\boldsymbol{\varepsilon}_{j,t}^\top \boldsymbol{H}_t^{-1}\boldsymbol{\varepsilon}_{j,t}} \cdot \boldsymbol{M}_{t,2:p}^\top \boldsymbol{H}_t^{-1}\boldsymbol{\varepsilon}_{j,t}.
\end{aligned} \tag{79}$$

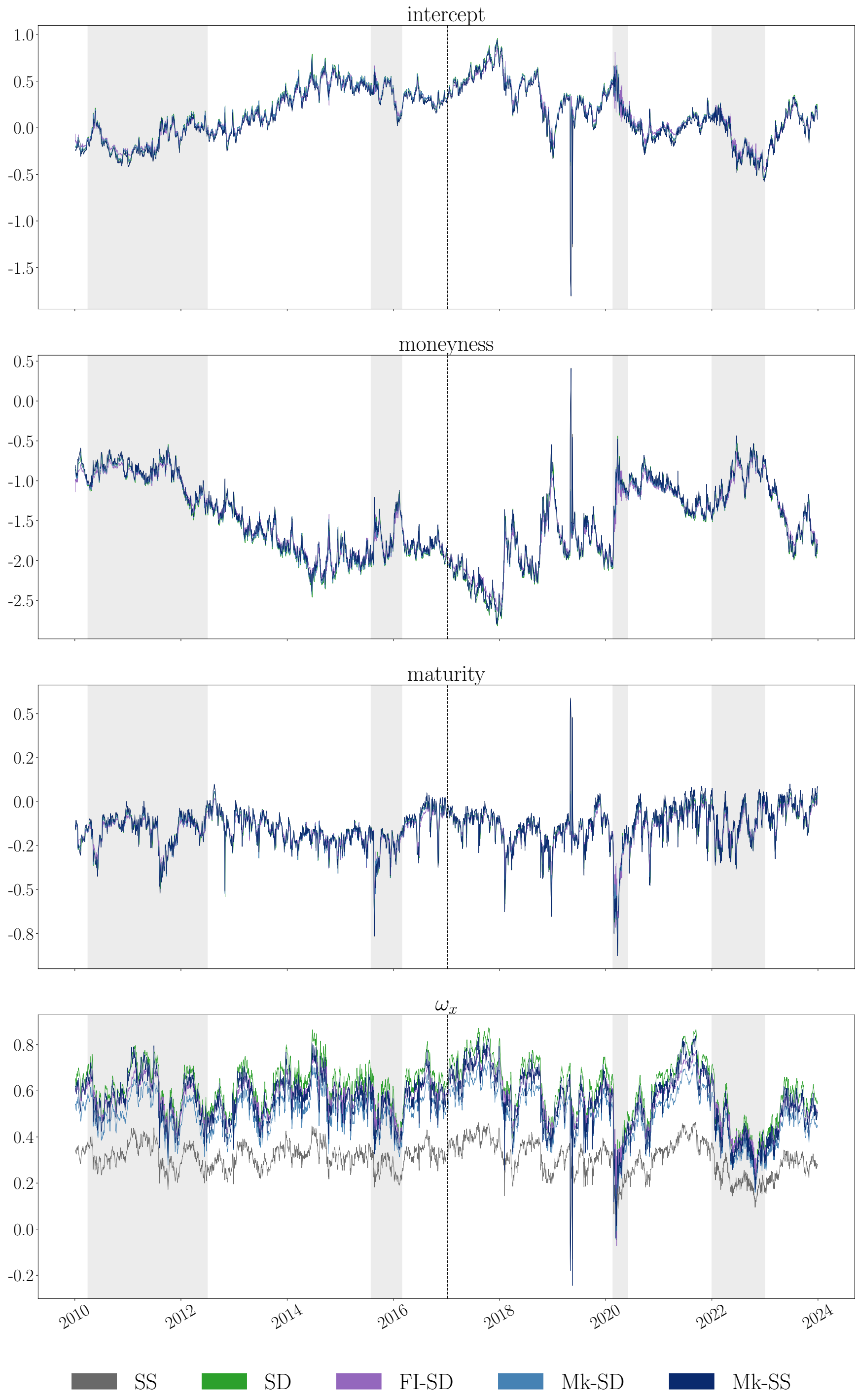
The Fisher information of $p_{\nu,j}$ with respect to $\tilde{\boldsymbol{\beta}}_t$ is state-independent since neither $\boldsymbol{M}_{t,2:p}$ nor $\boldsymbol{H}_t$ depend on $\boldsymbol{g}^{(j)}$:

$$\mathcal{I}_{\tilde{\boldsymbol{\beta}}} = \frac{\nu}{\nu-2} \cdot \frac{\nu+N_t}{\nu+N_t+2} \cdot \boldsymbol{M}_{t,2:p}^\top \boldsymbol{H}_t^{-1} \boldsymbol{M}_{t,2:p}. \tag{80}$$

$$
\begin{aligned}
[\boldsymbol{A}\boldsymbol{s}_t]_{2:p} &= \boldsymbol{A}\mathcal{I}_{\tilde{\boldsymbol{\beta}}}^{-1}\boldsymbol{\nabla}_t \\
&= \boldsymbol{A}\frac{(\nu-2)(\nu+N_t+2)}{\nu(\nu+N_t)}\left(\boldsymbol{M}_{t,2:p}^{\top}\boldsymbol{H}_t^{-1}\boldsymbol{M}_{t,2:p}\right)^{-1}\cdot \\
&\quad\cdot\sum_j \Pi_t^{(j)}\cdot\frac{\nu+N_t}{\nu-2+\boldsymbol{\varepsilon}_{j,t}^{\top}\boldsymbol{H}_t^{-1}\boldsymbol{\varepsilon}_{j,t}}\cdot\boldsymbol{M}_{t,2:p}^{\top}\boldsymbol{H}_t^{-1}\boldsymbol{\varepsilon}_{j,t} \qquad (81)\\
&= \boldsymbol{A}\left(\boldsymbol{M}_{t,2:p}^{\top}\boldsymbol{H}_t^{-1}\boldsymbol{M}_{t,2:p}\right)^{-1}\boldsymbol{M}_{t,2:p}^{\top}\boldsymbol{H}_t^{-1} \\
&\quad\cdot\sum_j \Pi_t^{(j)}\cdot\frac{1+(N_t+2)/\nu}{1+\boldsymbol{\varepsilon}_{j,t}^{\top}\boldsymbol{H}_t^{-1}\boldsymbol{\varepsilon}_{j,t}/(\nu-2)}\boldsymbol{\varepsilon}_{j,t}
\end{aligned}
$$

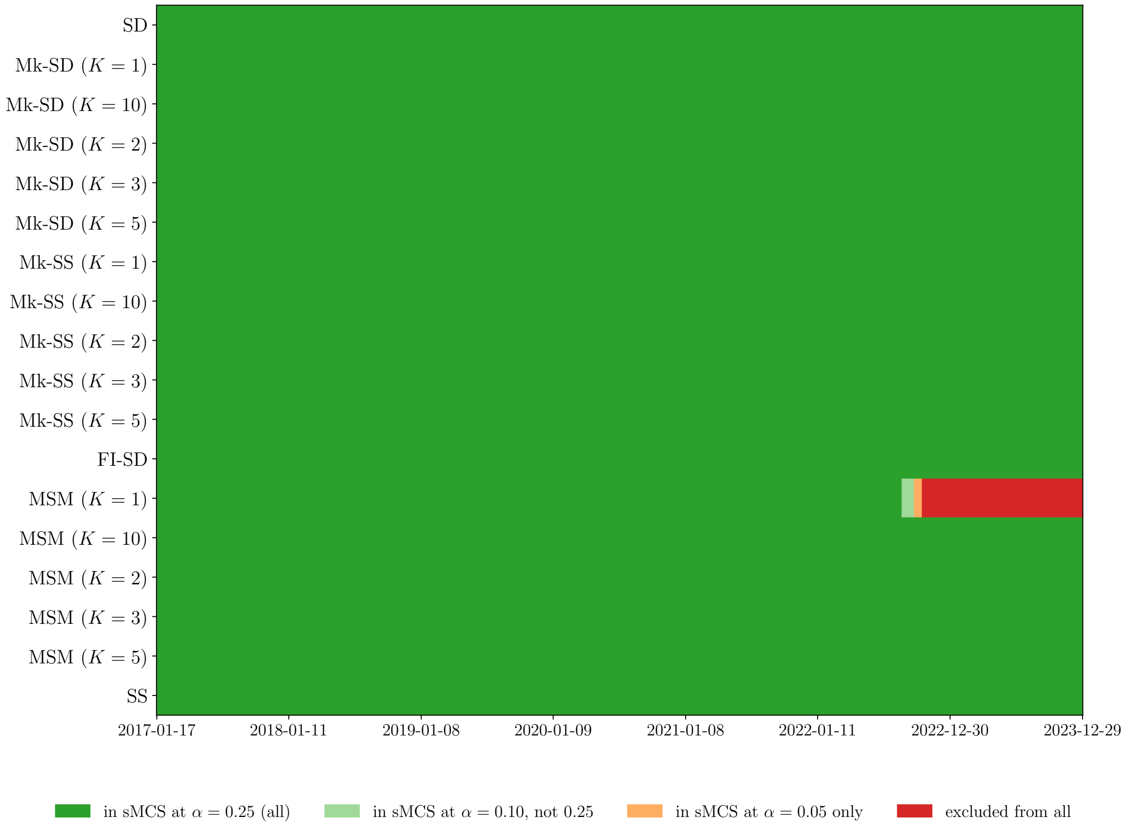
where the last line follows by canceling $(\nu + N_t)$ and dividing numerator and denominator by $(\nu - 2)$.

Figure 6: $\boldsymbol{\beta}_t$ over time



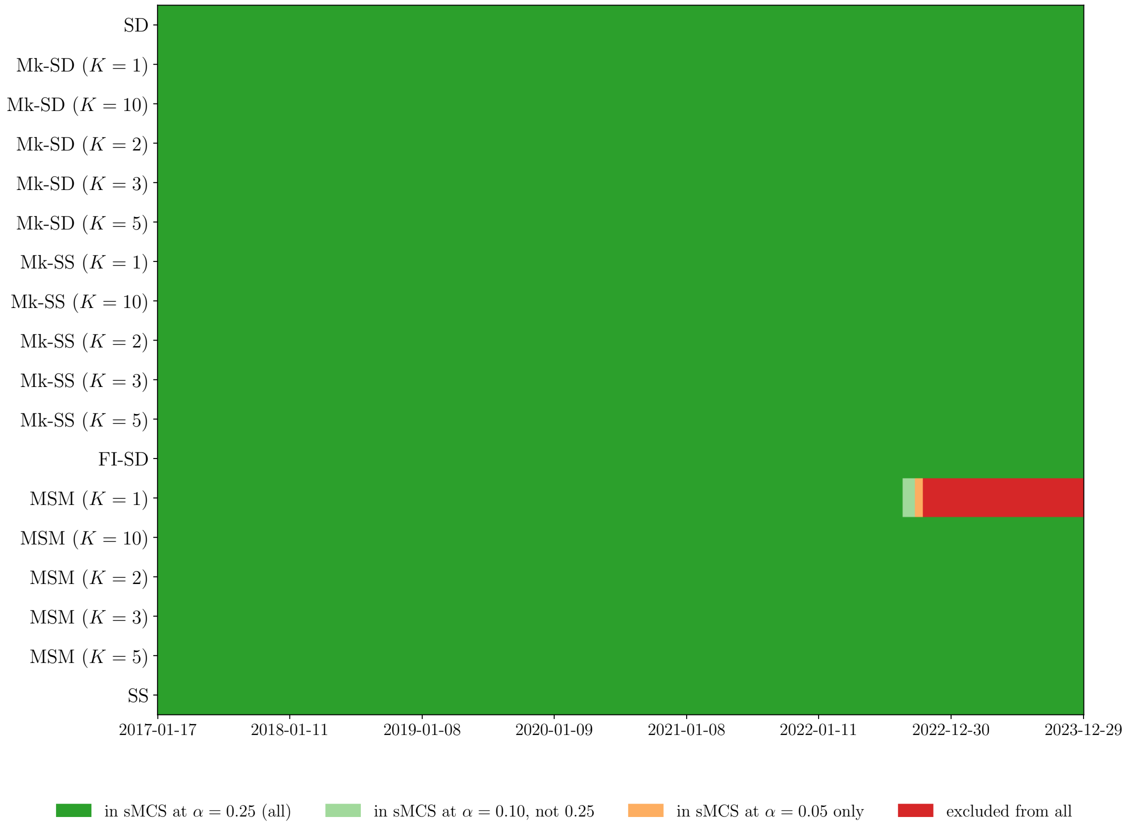
Note: Black dashed line denotes split between insample and out of sample period. Background gray areas denote crises period, including - 2010-04 to 2012-07 Euro debt crisis - 2015-08 to 2016-02 — China/EM selloff - 2020-02 to 2020-06 — COVID crash - 2022-01 to 2022-12 — 2022 rate hike bear market.

Figure 7: 5-steps ahead MSE Model Confidence Sequence Inclusions



Note: Sequential model confidence sets (sMCS) under the weak hypothesis. At each time t, model i is included if there is no sufficient evidence that its cumulative average conditional loss exceeds any competitor's. Colors indicate membership at different Confidence Levels (α): dark green = in sMCS at all three levels ($\alpha \leq 0.25$); light green = $\alpha \leq 0.10$; orange = $\alpha \leq 0.05$ only; red = excluded from all. The procedure provides a time-uniform family-wise error guarantee simultaneously over all t; models may re-enter the set as cumulative evidence evolves. E-processes use a fixed per-pair betting parameter $\lambda_{ij} = (2c_{ij})^{-1}$ with $c_{ij} = 2\max_t |d_{ij,t}|$ and $d_{ij,t}$ the MSE differential between model i and j at time t.

Figure 8: 22-steps ahead MSE Model Confidence Sequence Inclusions



Note: Sequential model confidence sets (sMCS) under the weak hypothesis. At each time t, model i is included if there is no sufficient evidence that its cumulative average conditional loss exceeds any competitor's. Colors indicate membership at different Confidence Levels (α): dark green = in sMCS at all three levels ($\alpha \leq 0.25$); light green = $\alpha \leq 0.10$; orange = $\alpha \leq 0.05$ only; red = excluded from all. The procedure provides a time-uniform family-wise error guarantee simultaneously over all t; models may re-enter the set as cumulative evidence evolves. E-processes use a fixed per-pair betting parameter $\lambda_{ij} = (2c_{ij})^{-1}$ with $c_{ij} = 2\max_t |d_{ij,t}|$ and $d_{ij,t}$ the MSE differential between model i and j at time t.